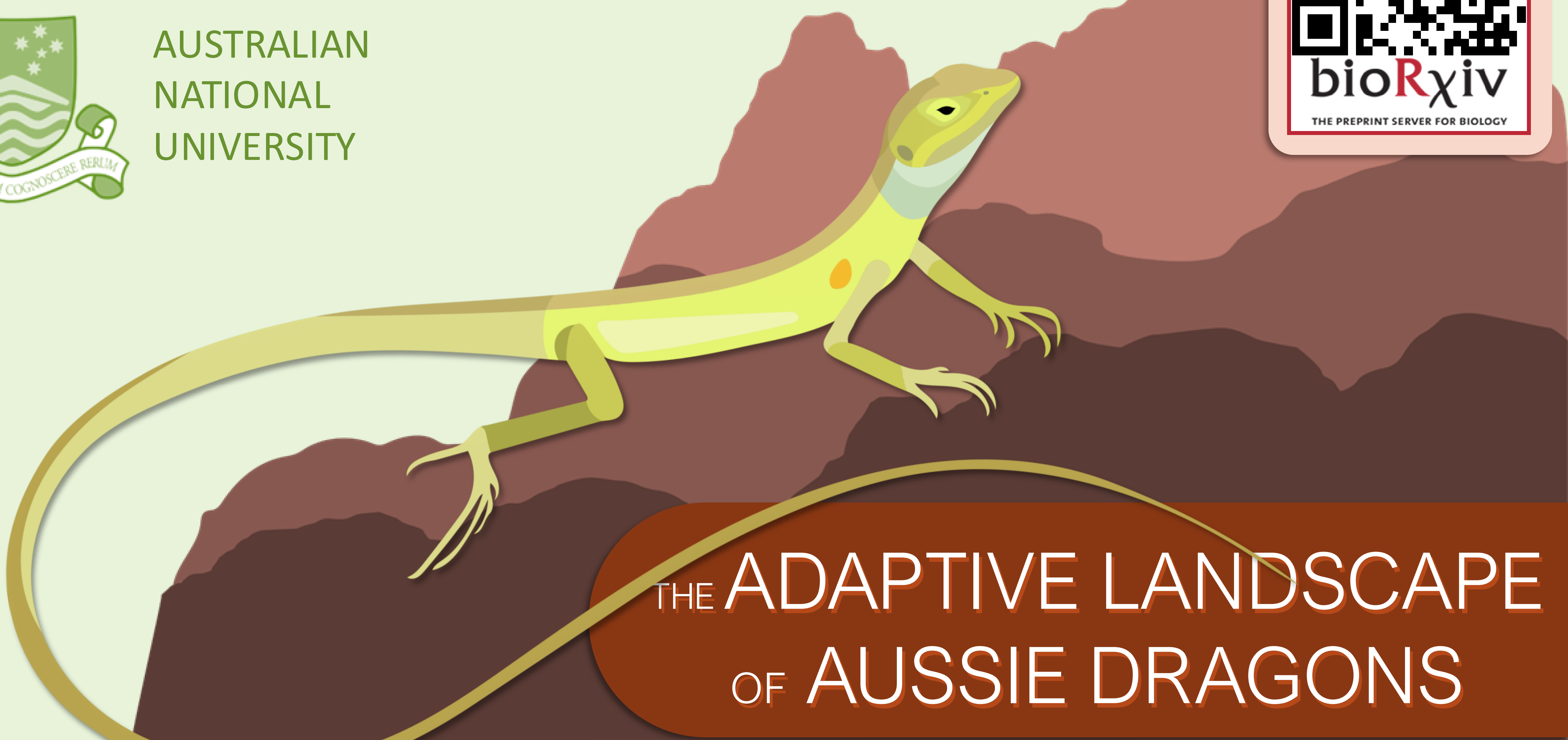
Ian Brennan

AUSTRALIAN NATIONAL UNIVERSITY



THE ADAPTIVE LANDSCAPE OF AUSSIE DRAGONS

Ian Brennan

AUSTRALIAN
NATIONAL
UNIVERSITY

COLLABORATORS:

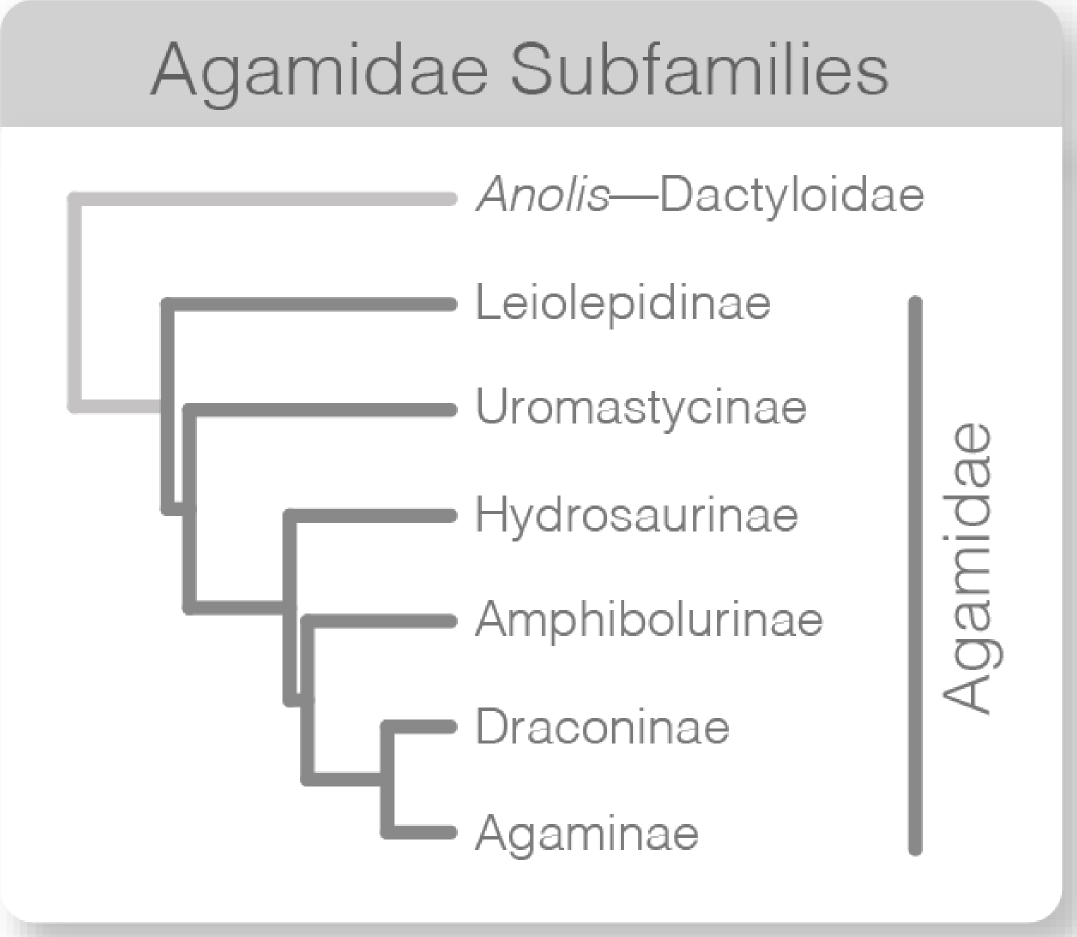
JANE MELVILLE	MV
NATALIE COOPER	NHM
JOANNA SUMNER	MV
LEO TEDESCHI	CSIRO
LIZ BROADY	CSIRO
SCOTT KEOGH	ANU

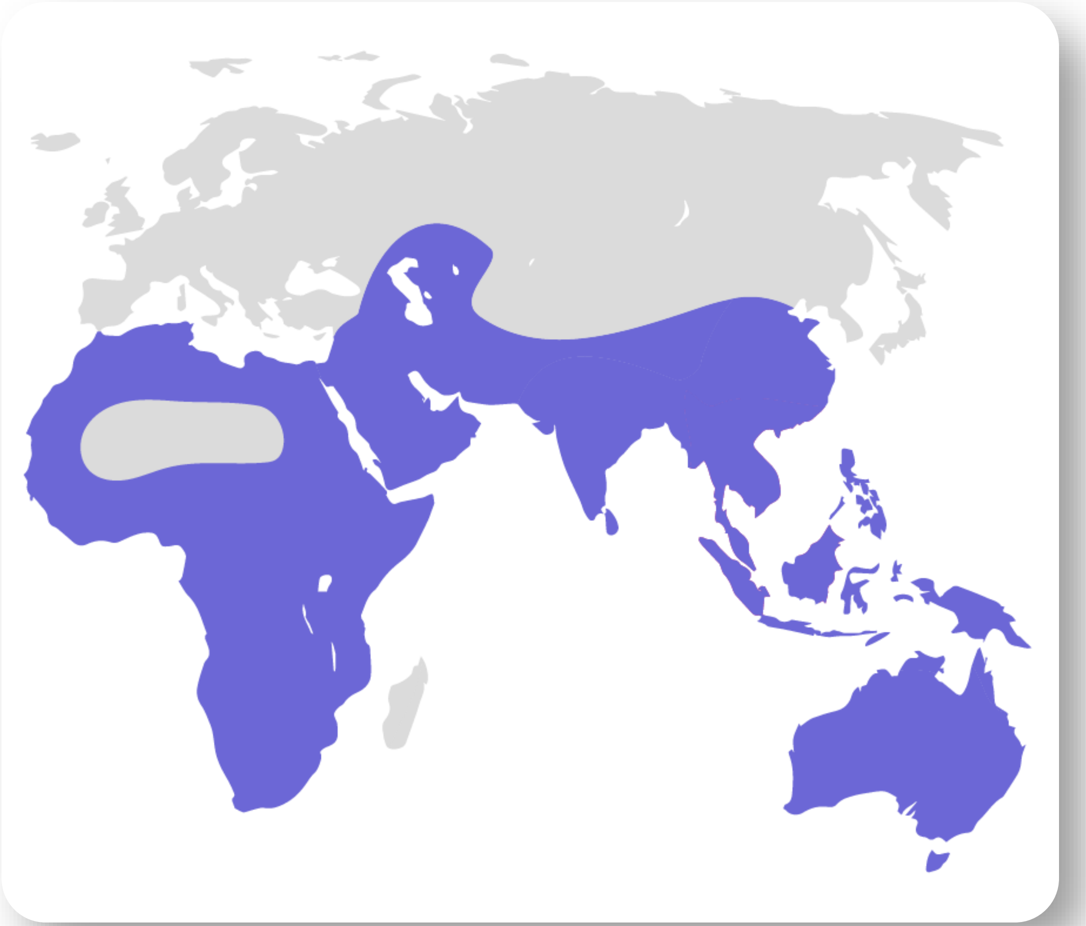
MARIE
SKŁODOWSKA
CURIE
ACTIONS

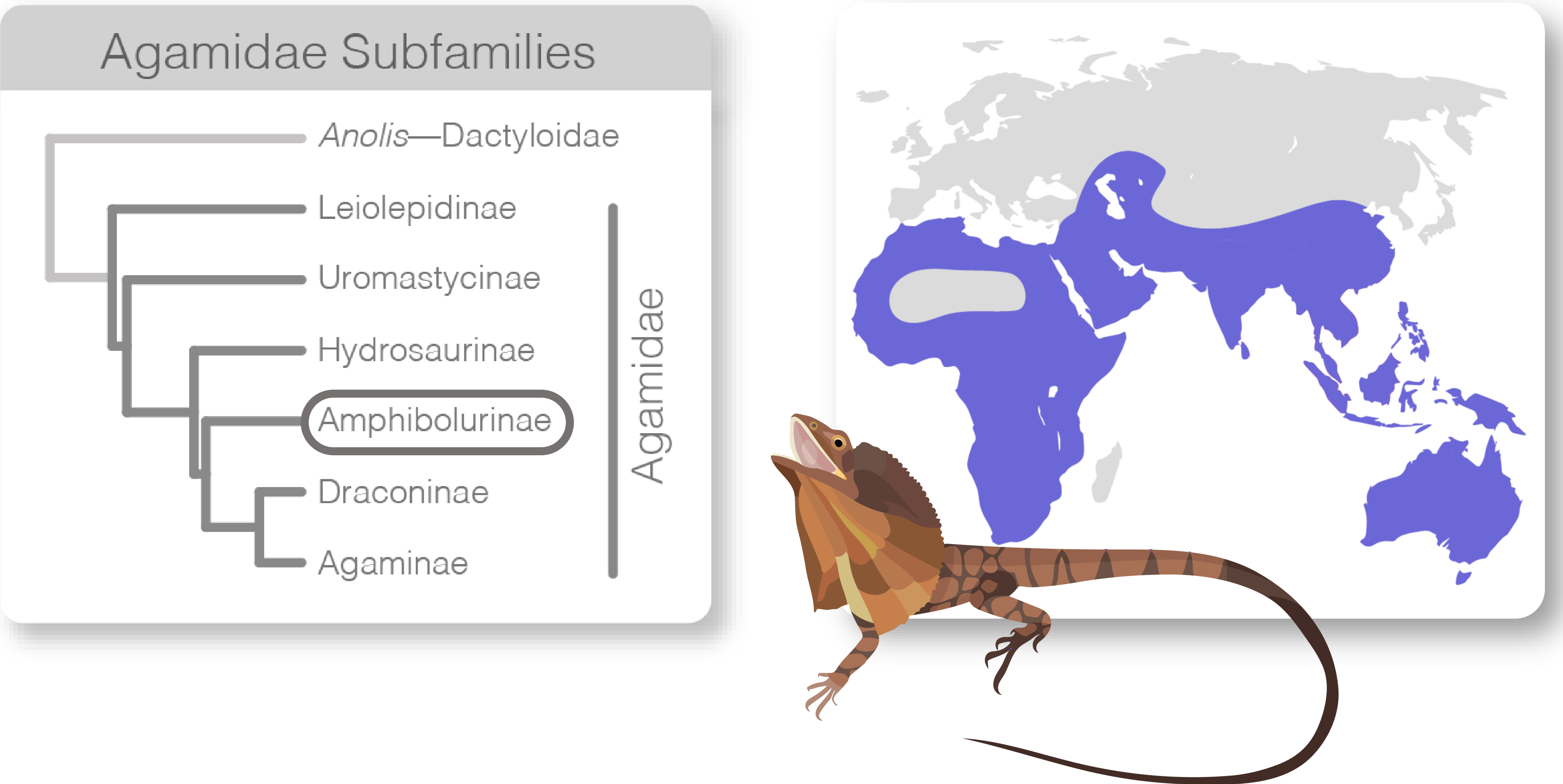
BIOPLATFORMS
AUSTRALIA

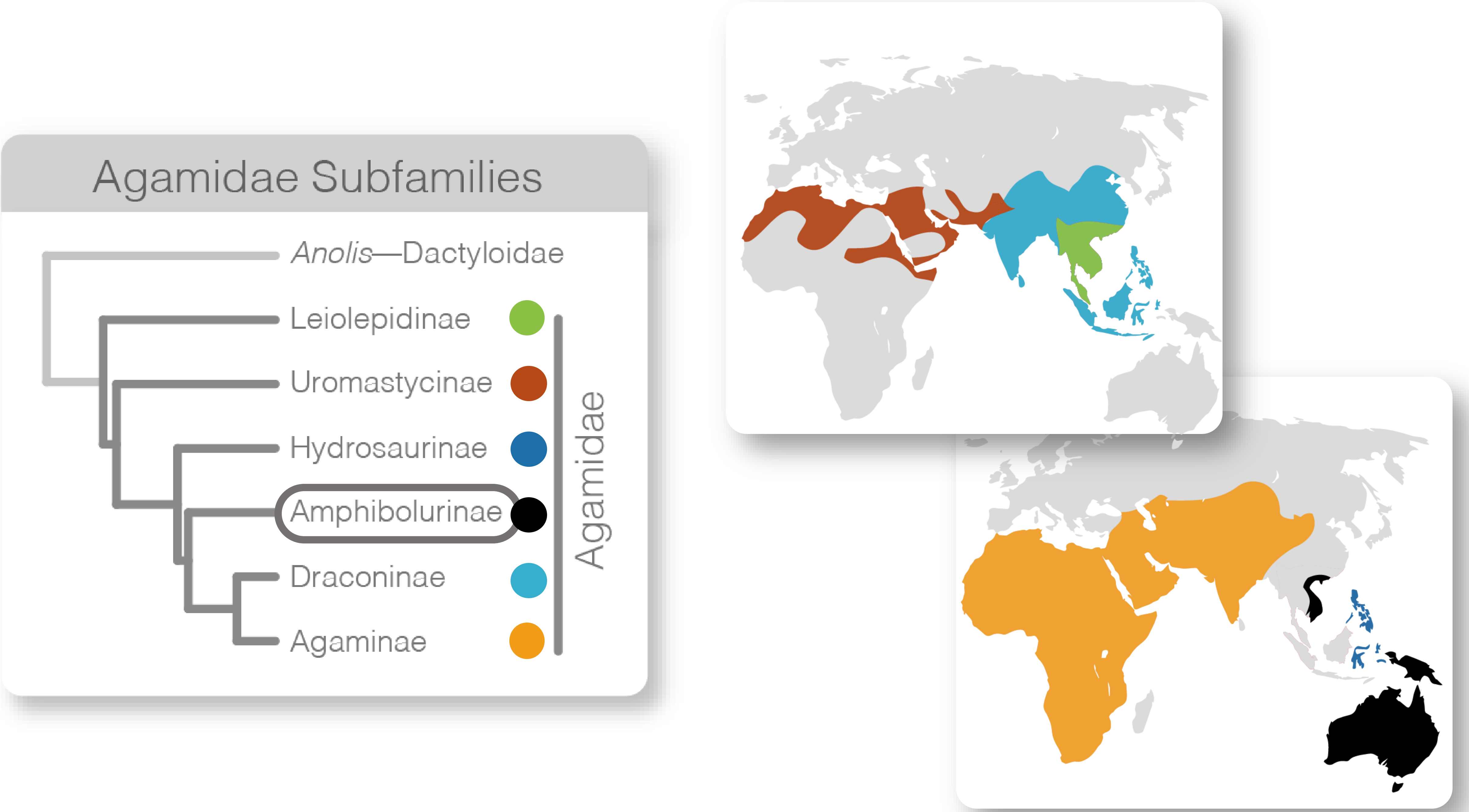
AGAMIDAE
DRAGON
LIZARDS

ICONIC
AUSTRALIAN
REPTILES


Agamidae Subfamilies
Anolis—Dactyloidae
Leiolepidinae
Uromastycinae
Hydrosaurinae
Amphibolurinae
Draconinae
Agaminae
Agamidae




Agamidae Subfamilies
Anolis—Dactyloidae
Leiolepidinae
Uromastycinae
Hydrosaurinae
Amphibolurinae
Draconinae
Agaminae
Agamidae


Agamidae Subfamilies
Anolis—Dactyloidae
Leiolepidinae
Uromastycinae
Hydrosaurinae
Amphibolurinae
Draconinae
Agaminae
Agamidae

1

BUILD A TREE OF THE AMPHIBOLURINAE

1

BUILD A TREE OF THE AMPHIBOLURINAE

2

EXPLORE THEIR MORPHOLOGICAL EVOLUTION

AusARG PHYLOGENOMICS

BIOPLATFORMS AUSTRALIA

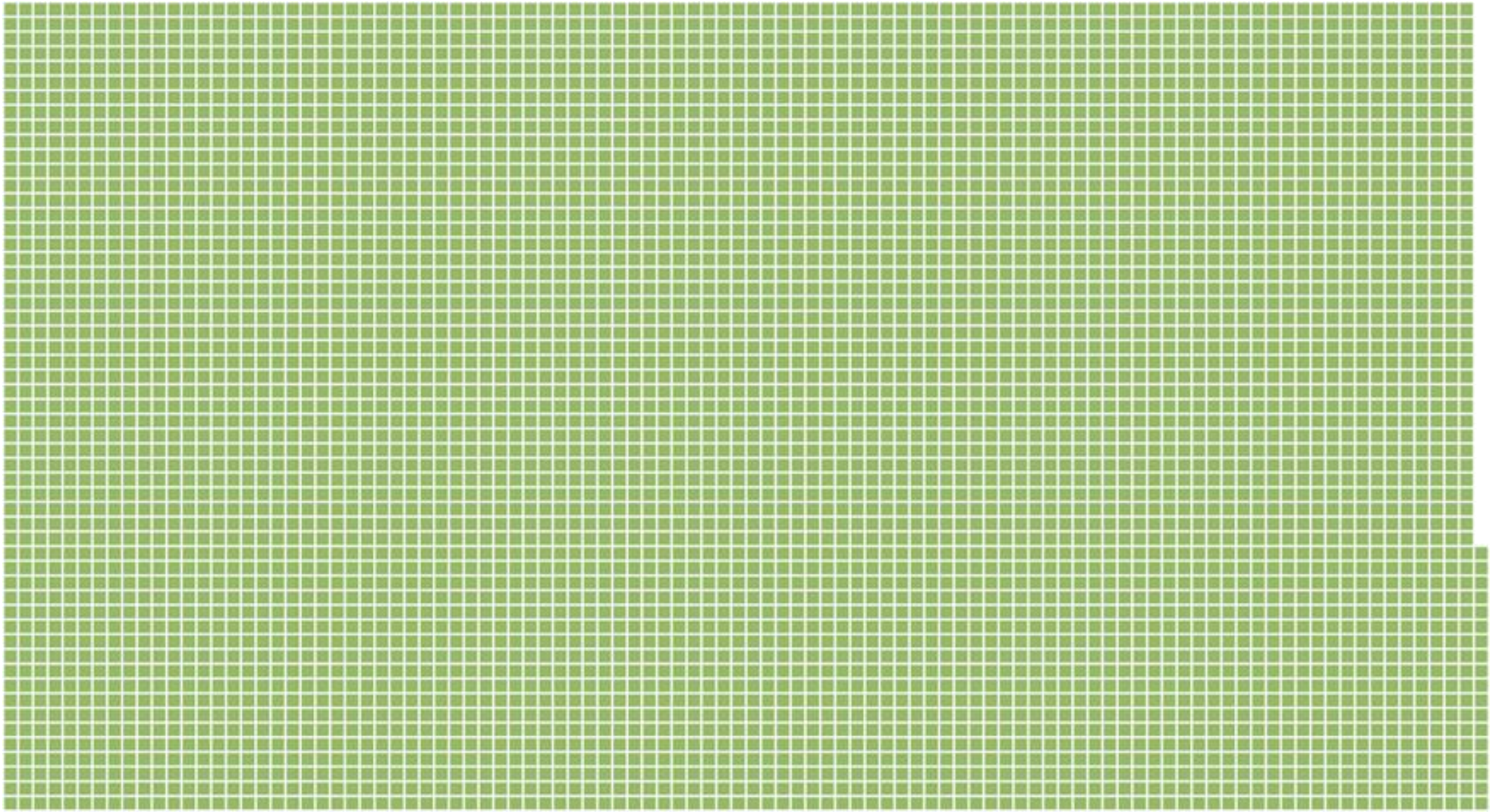
PIPESNAKE

AusARG PHYLOGENOMICS

BIOPLATFORMS AUSTRALIA

PIPESNAKE

= 1 LOCUS



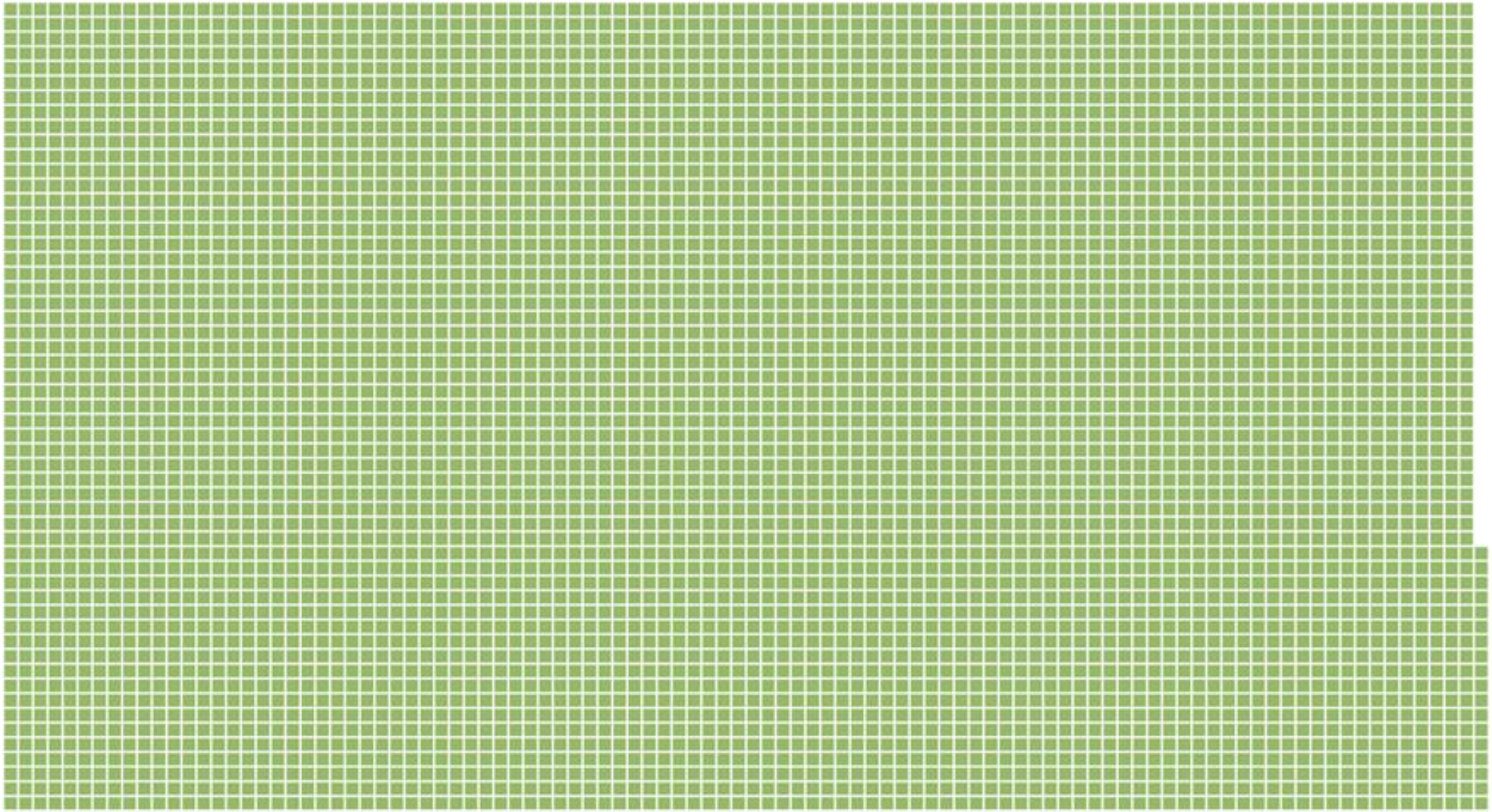
AusARG PHYLOGENOMICS

BIOPLATFORMS AUSTRALIA

PIPESNAKE

= 1 LOCUS

5,440 TOTAL LOCI



AusARG PHYLOGENOMICS

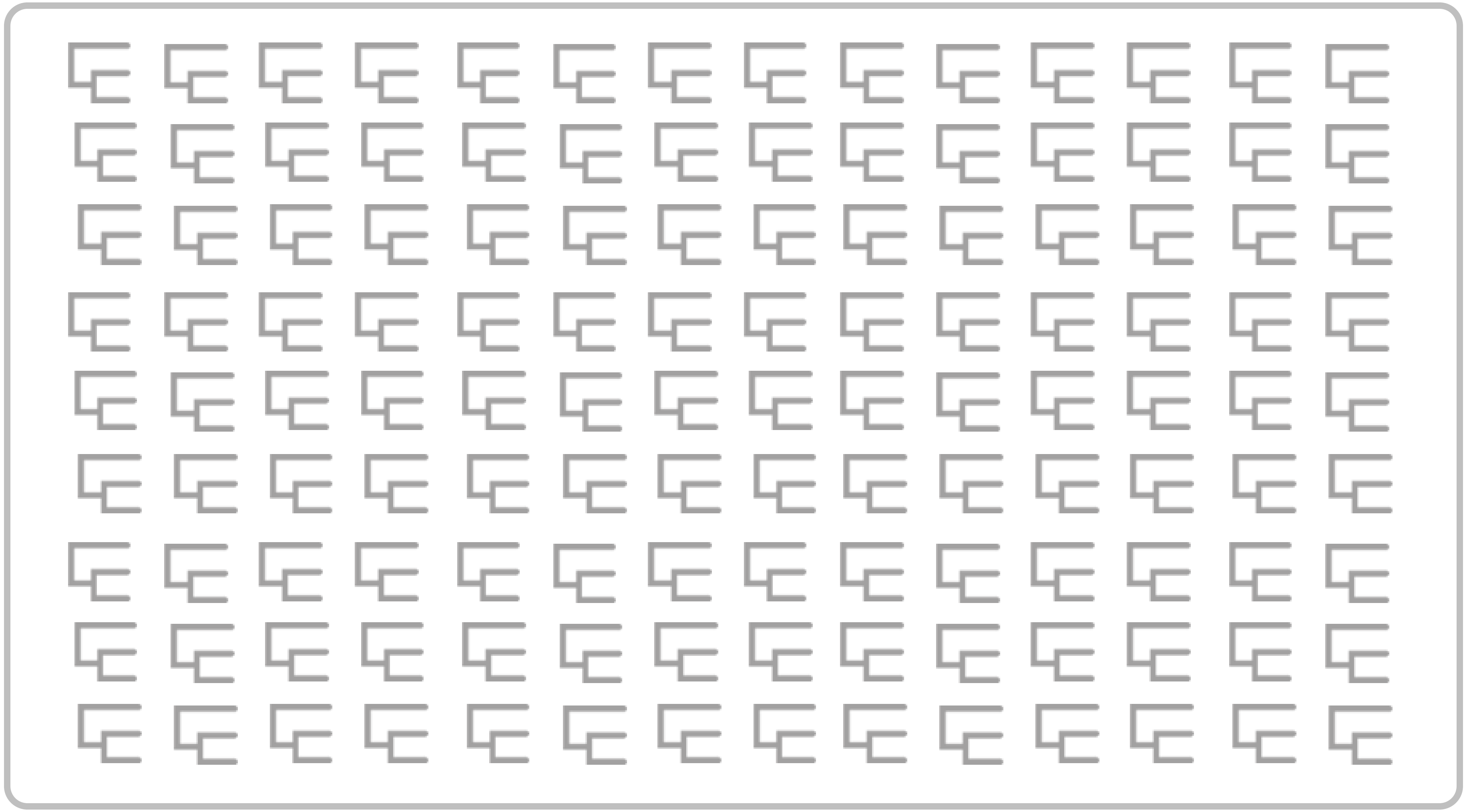
BIOPLATFORMS AUSTRALIA

PIPESNAKE

= 1 LOCUS

5,440 TOTAL LOCI

5,440 GENE TREES



AusARG PHYLOGENOMICS

BIOPLATFORMS AUSTRALIA

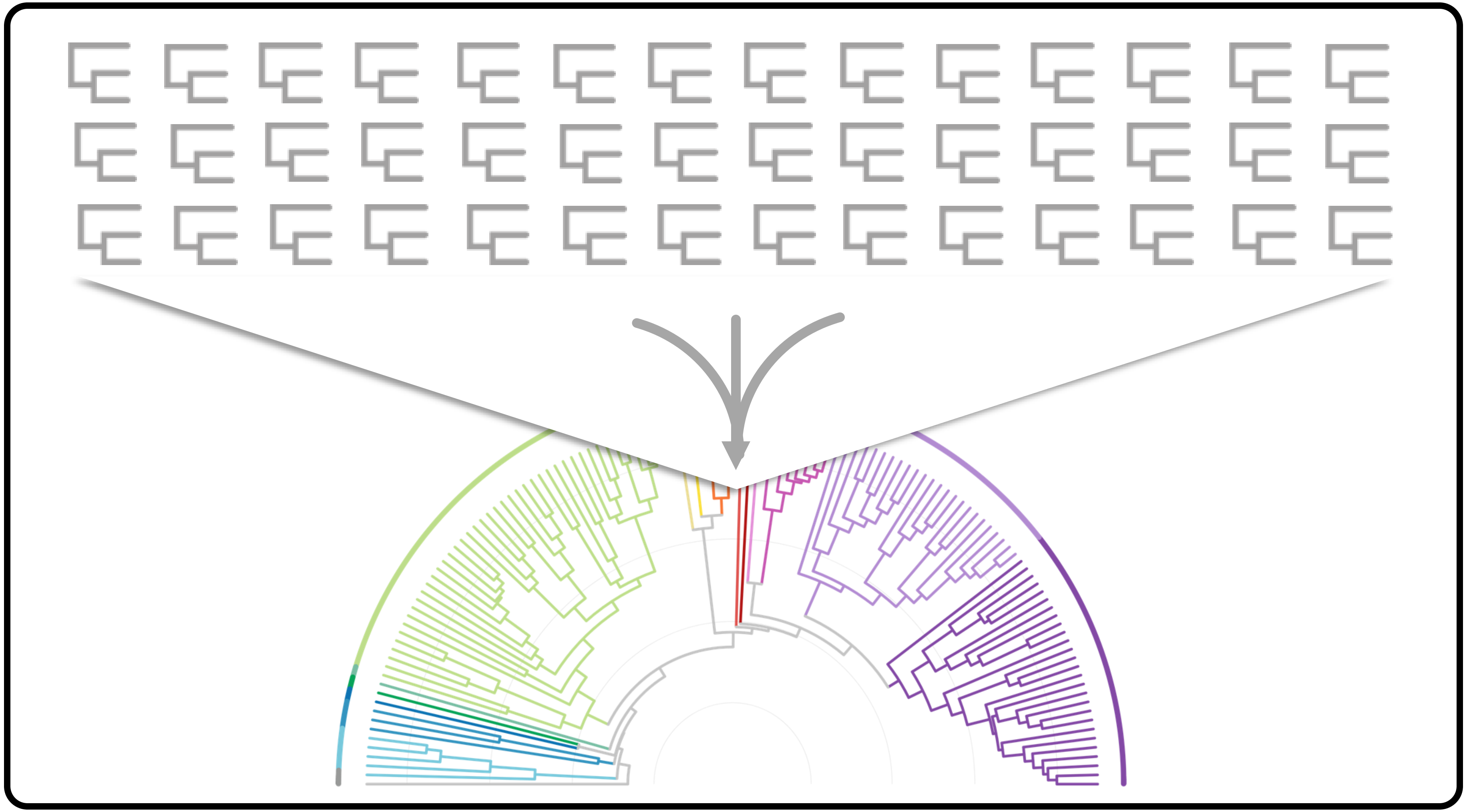
PIPESNAKE

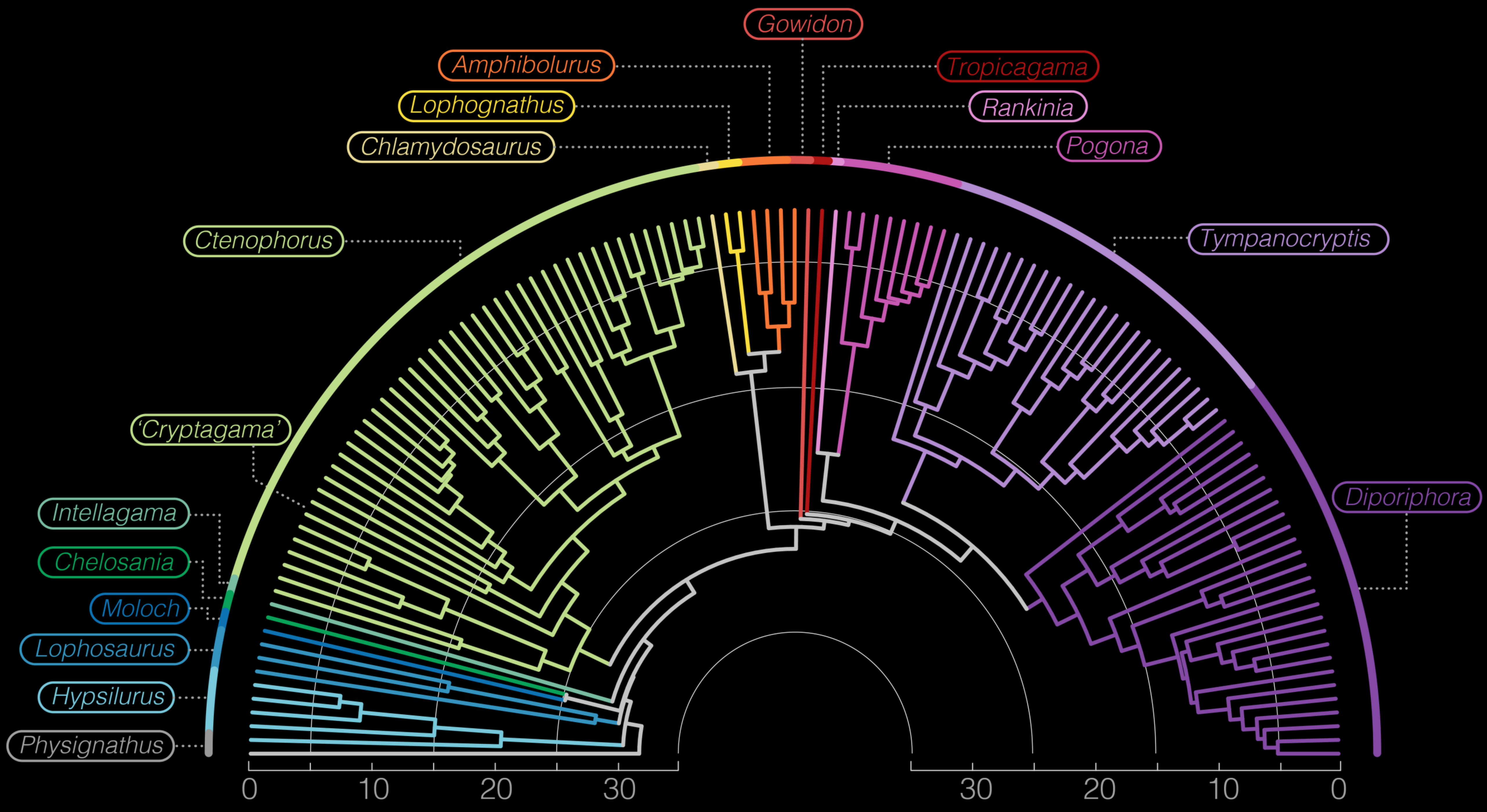
= 1 LOCUS

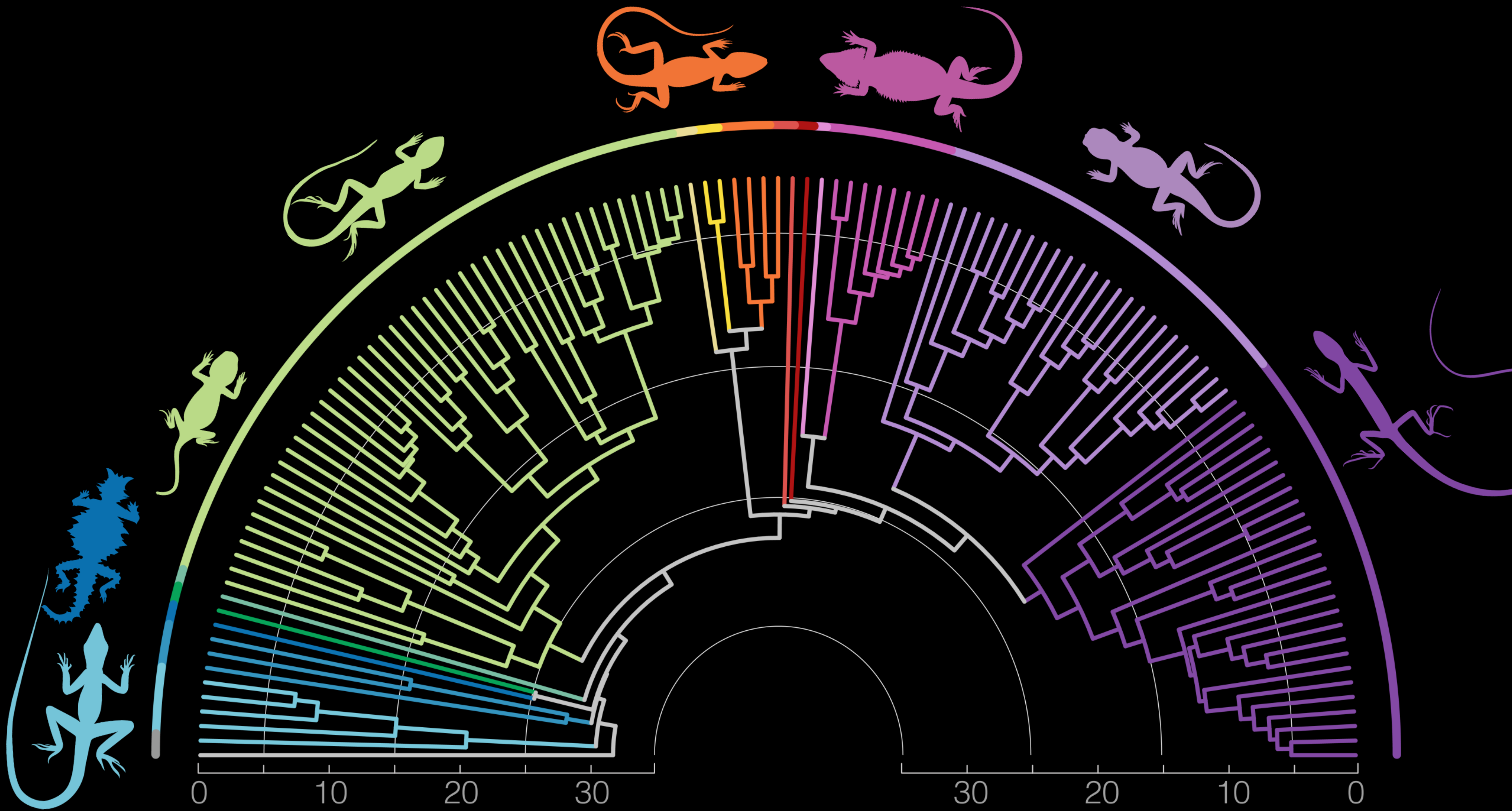
5,440 TOTAL LOCI

5,440 GENE TREES

1 SPECIES TREE

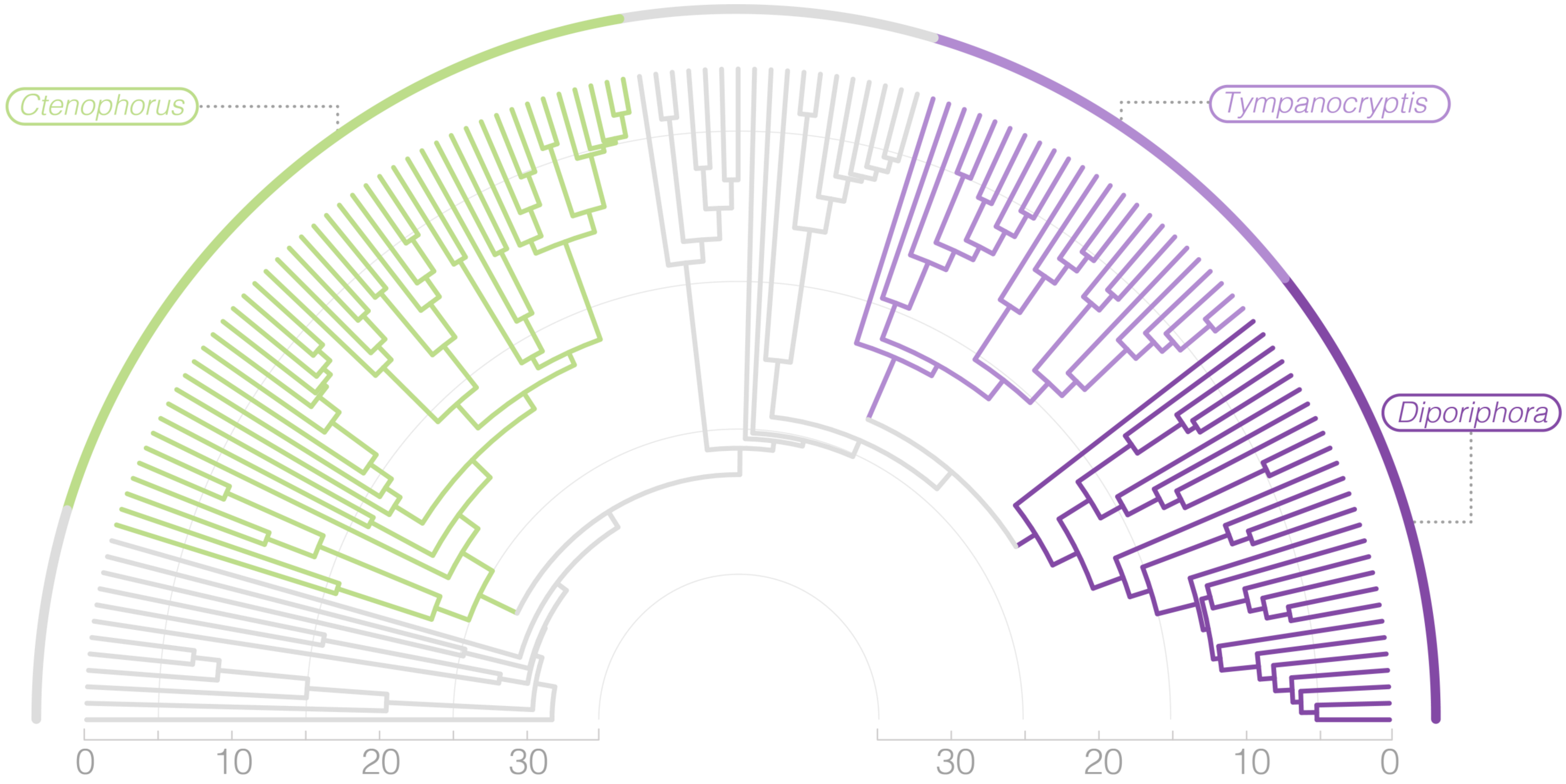



Gowidon
Amphibolurus
Tropicagama
Lophognathus
Rankinia
Chlamydosaurus
Pogona
Ctenophorus
Tympanocryptis
'Cryptagama'
Diporiphora
Intellagama
Chelosania
Moloch
Lophosaurus
Hypsilurus
Physignathus
0
10
20
30
30
20
10
0

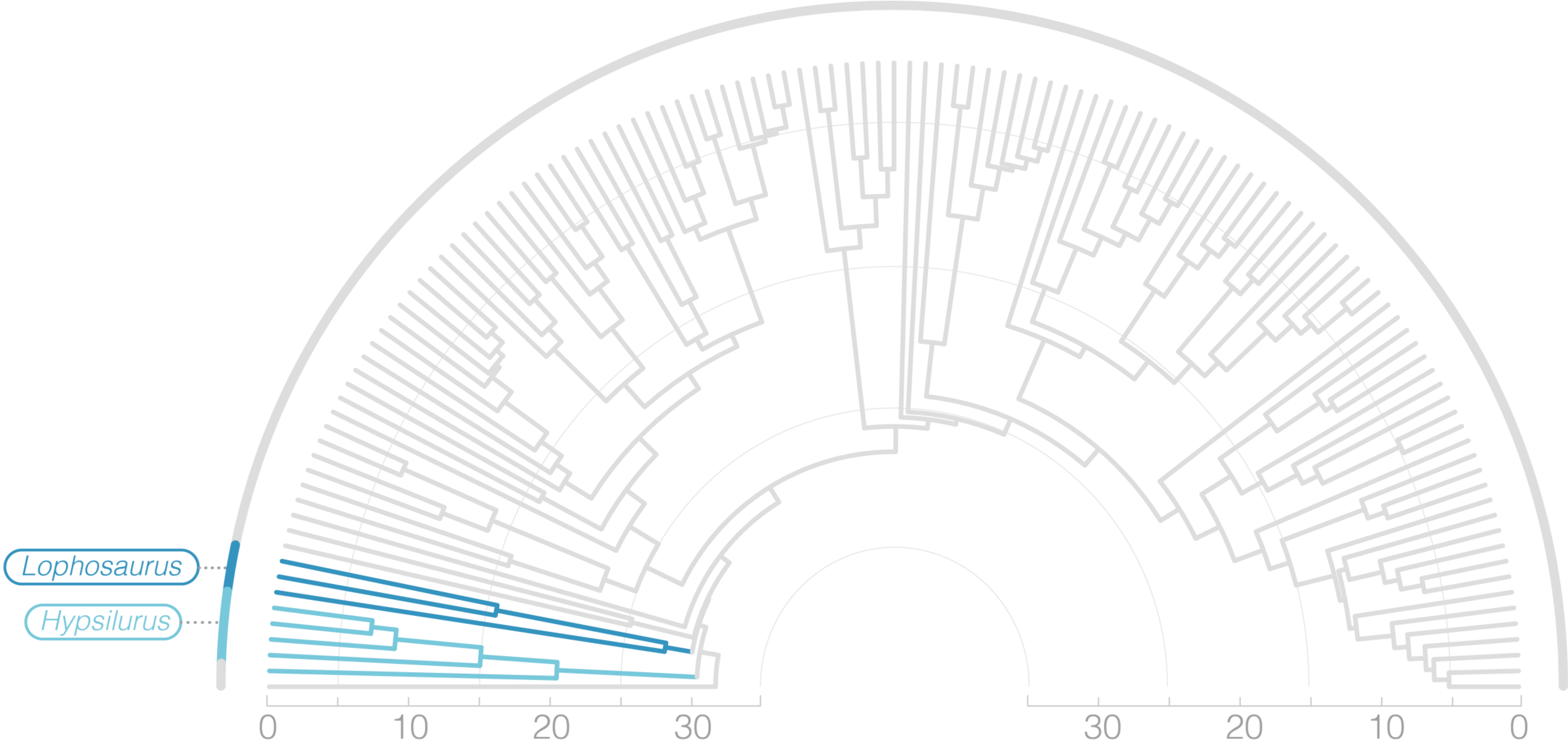


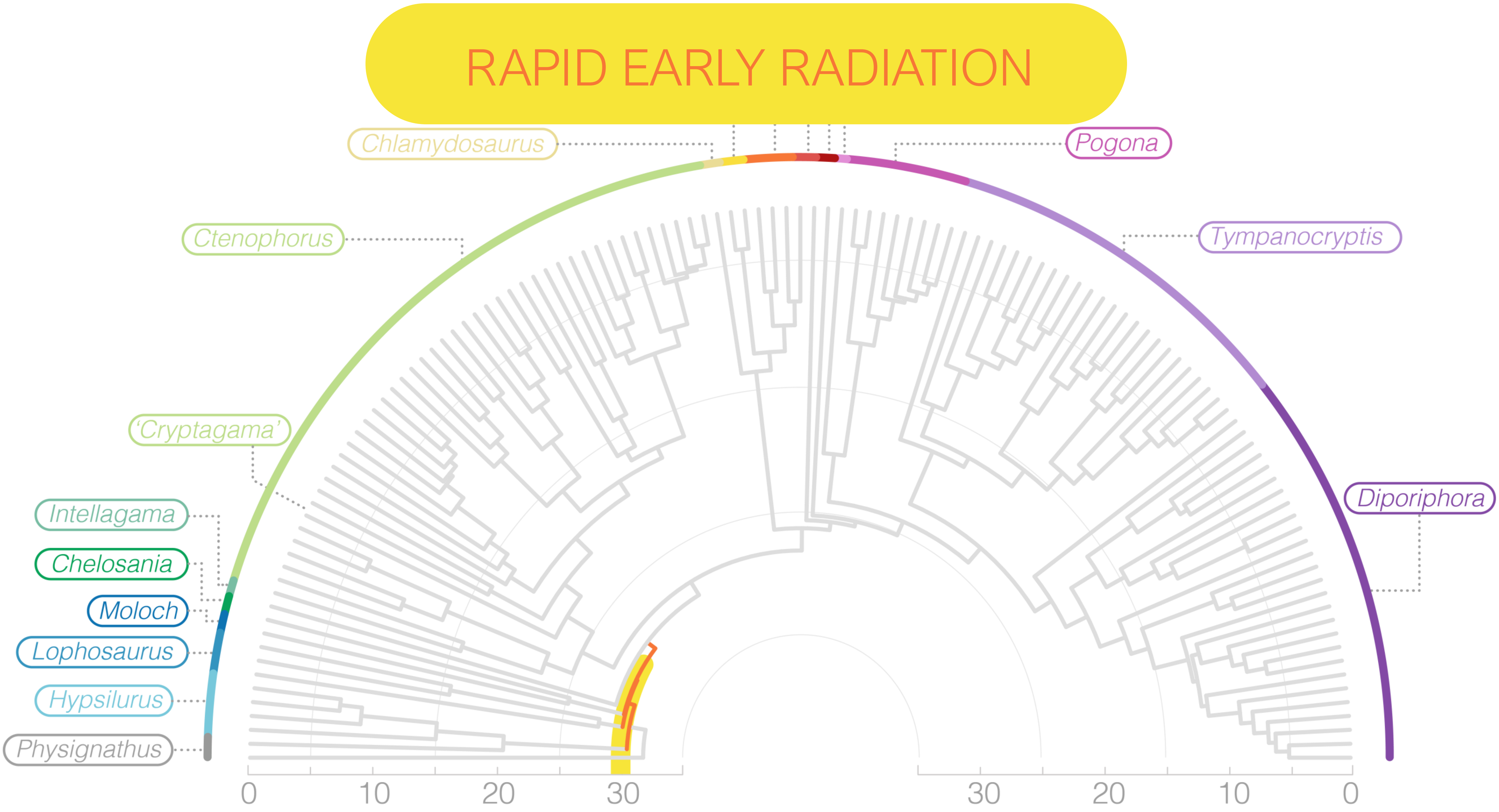
0
10
20
30
30
20
10
0

RICHNESS CONCENTRATED IN 3 GENERA

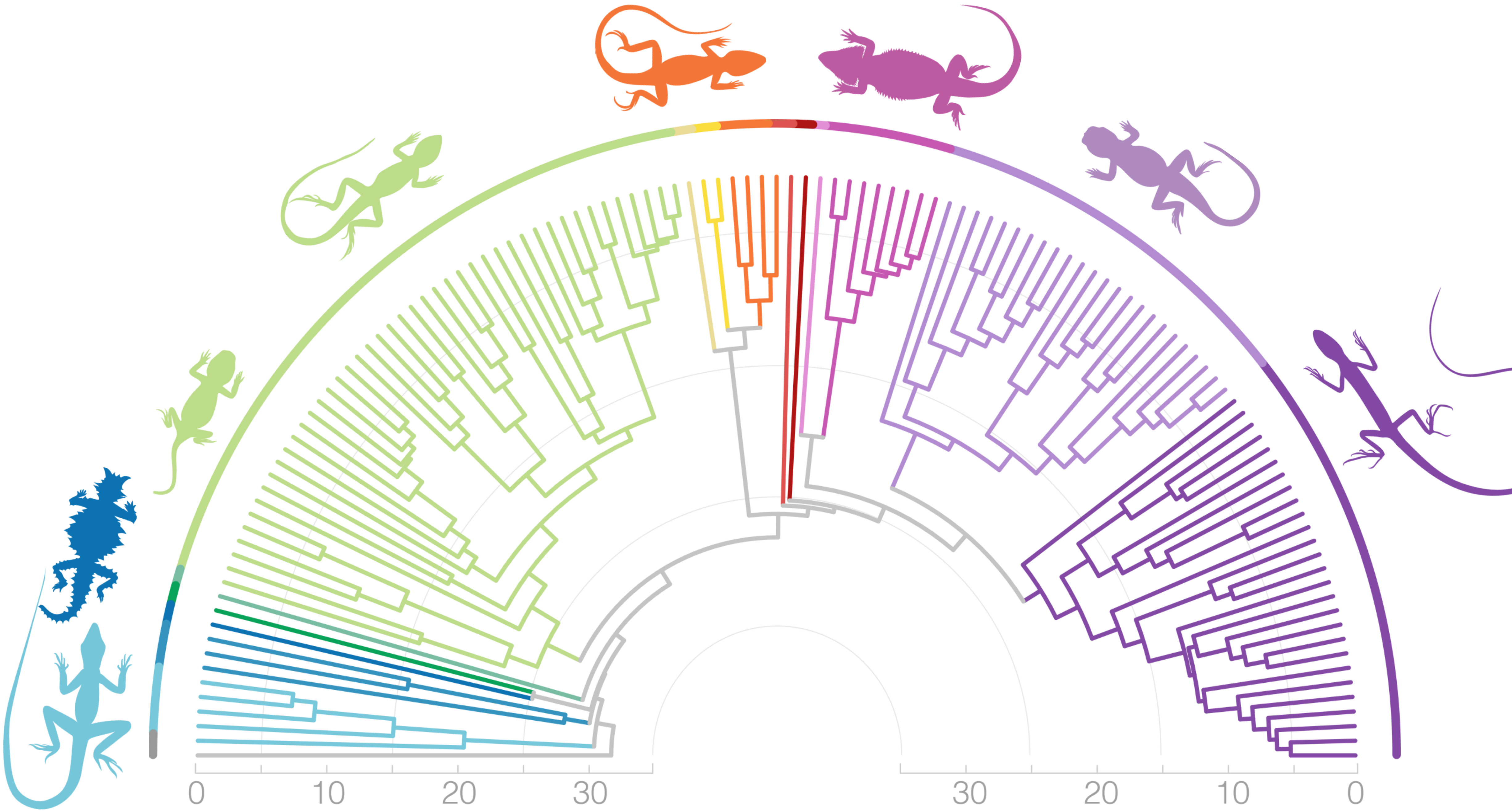


TREE DRAGONS ARE NOT A CLADE

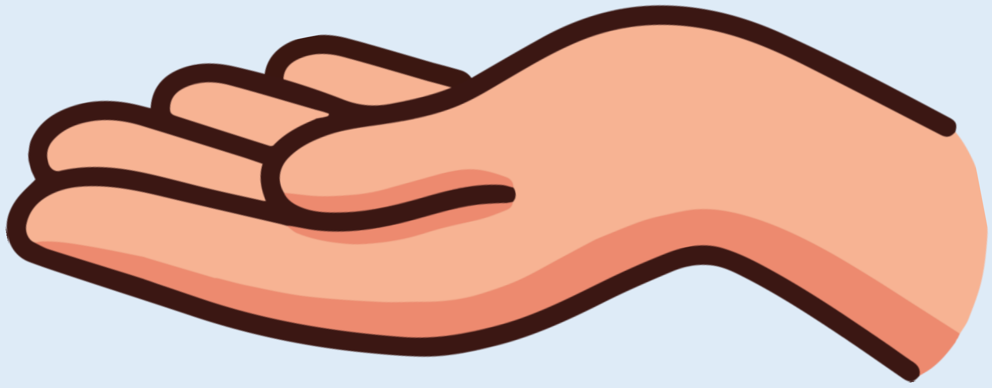


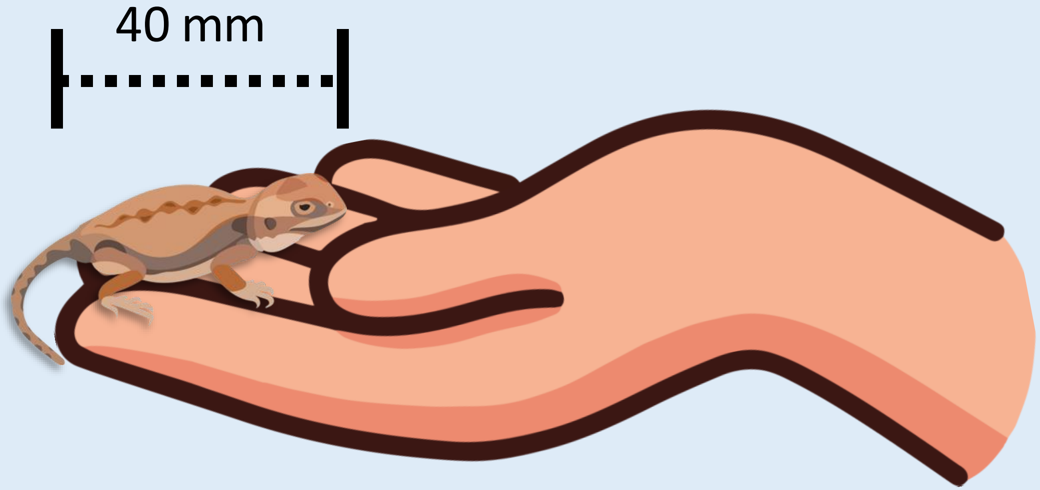


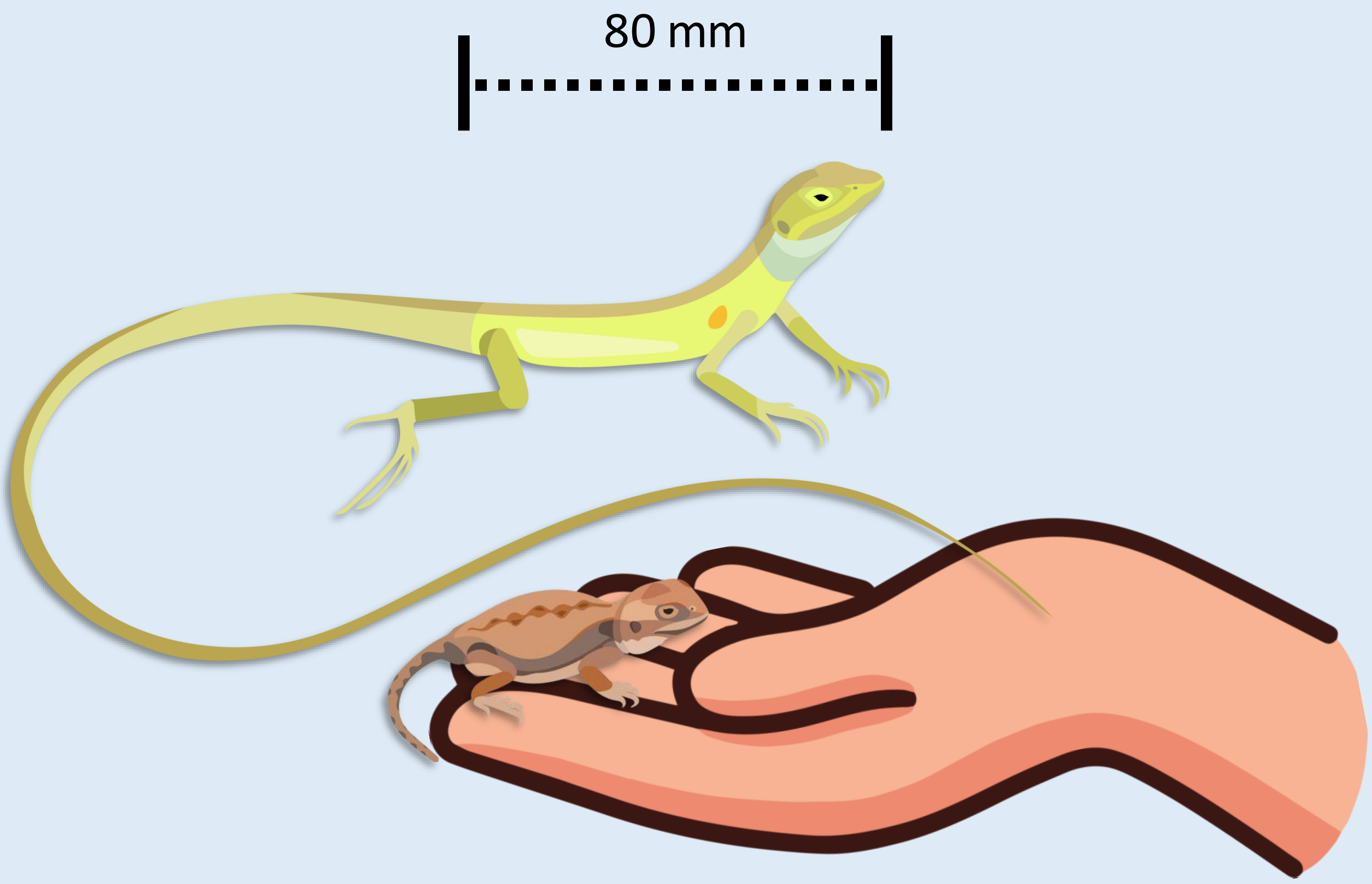
RAPID EARLY RADIATION
Chlamydosaurus
Pogona
Ctenophorus
Tympanocryptis
'Cryptagama'
Diporiphora
Intellagama
Chelosania
Moloch
Lophosaurus
Hypsilurus
Physignathus
0
10
20
30
30
20
10
0

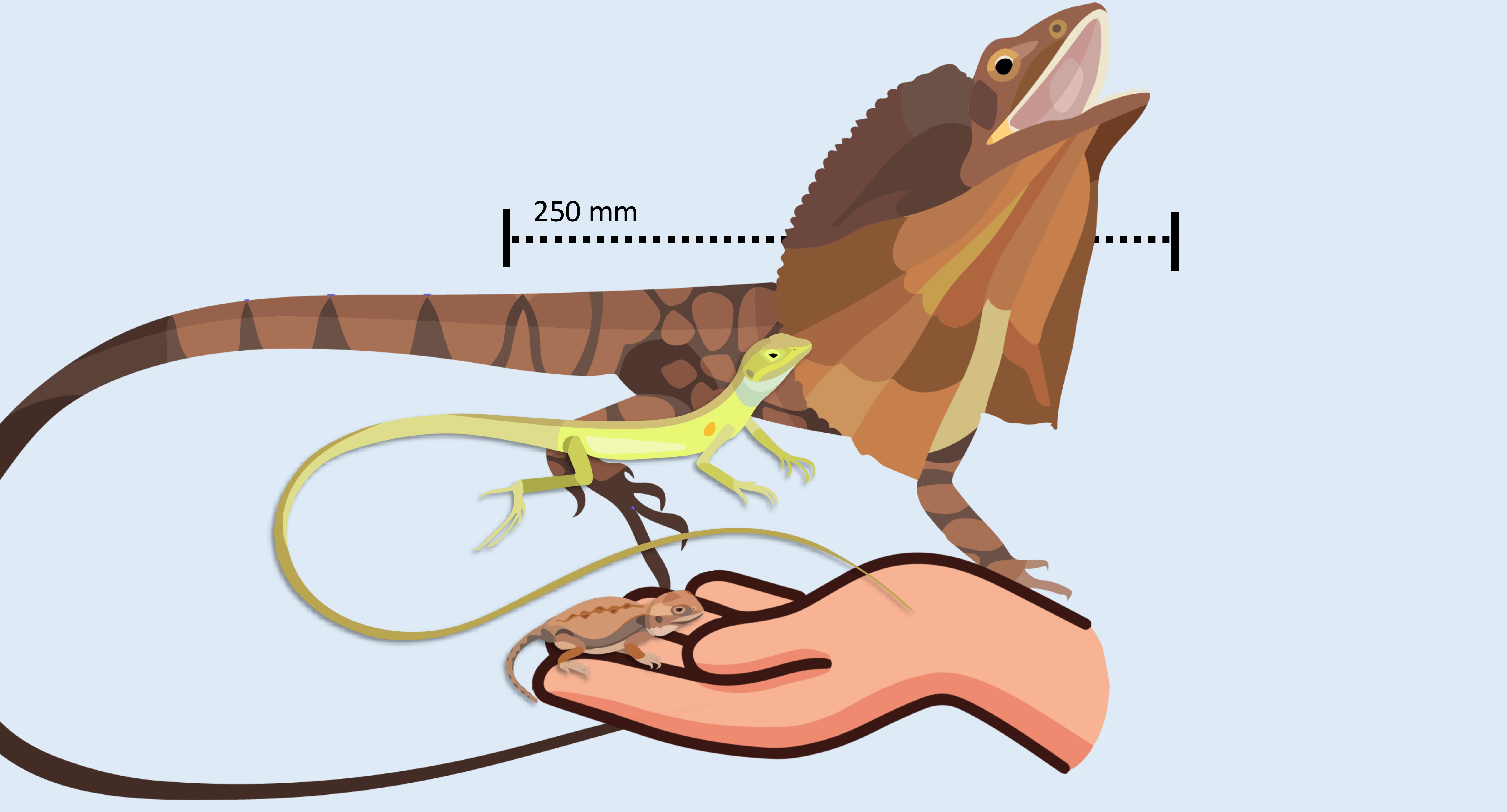


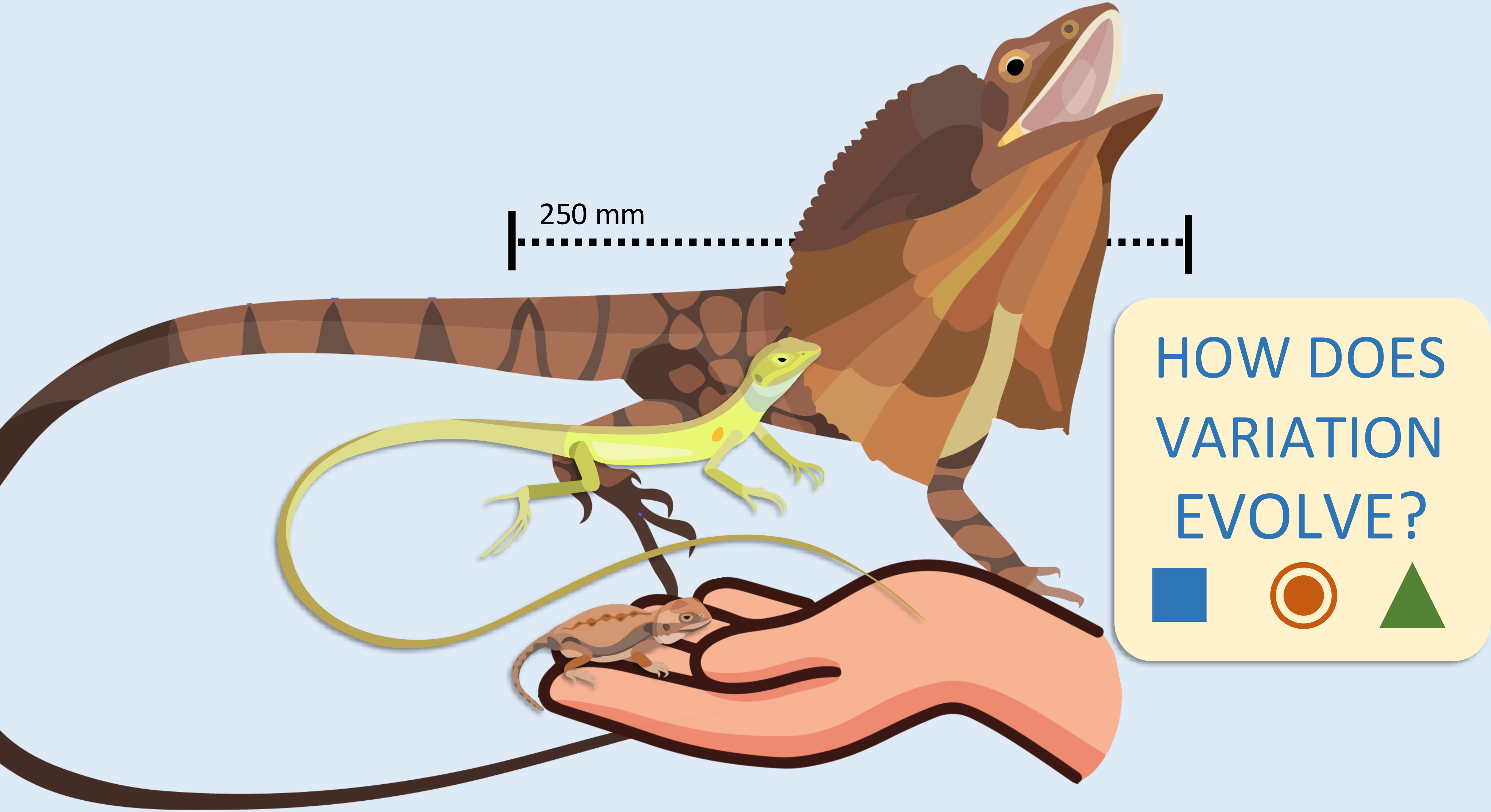
0
10
20
30
30
20
10
0




40 mm

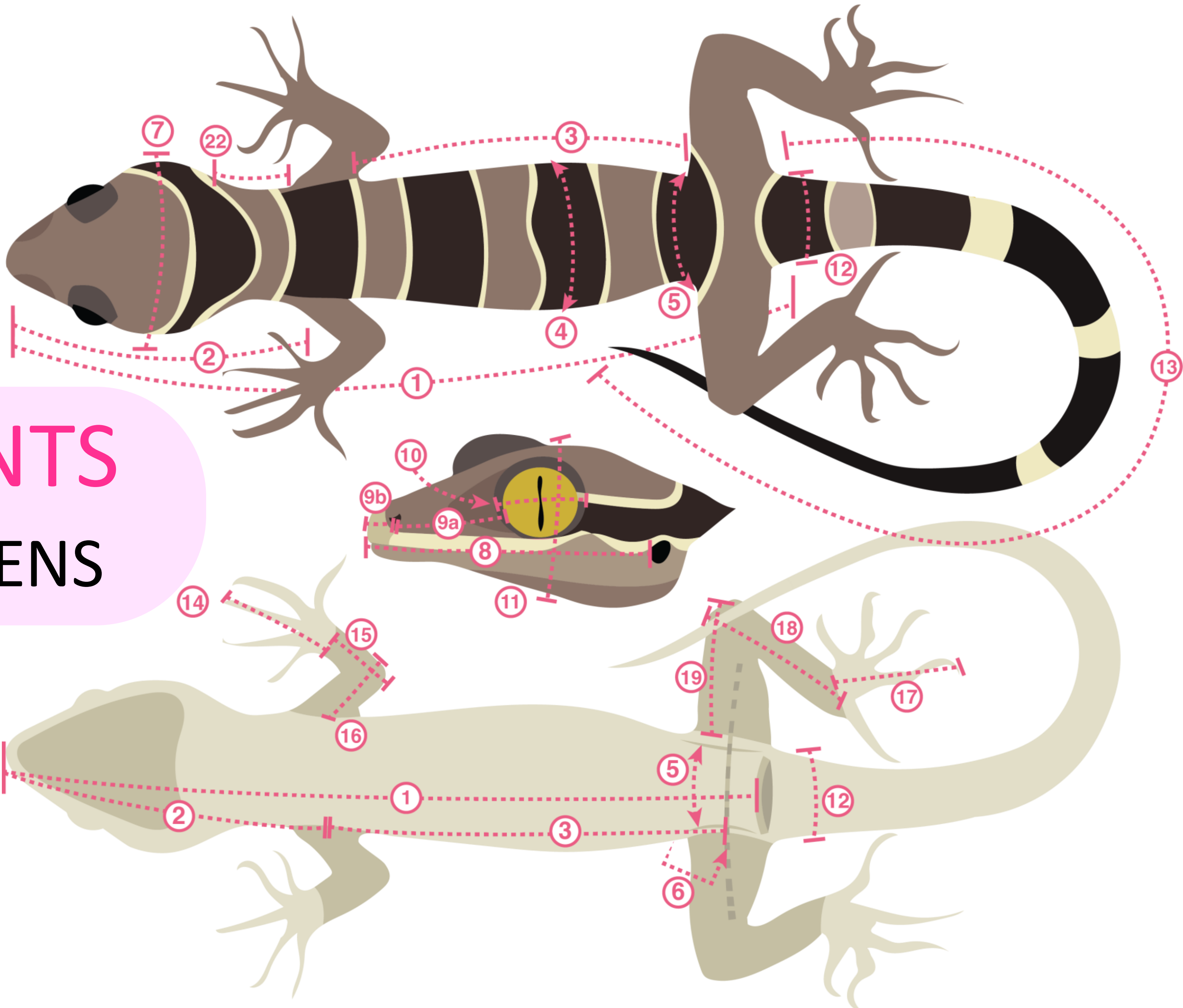

80 mm


250 mm


250 mm
HOW DOES
VARIATION
EVOLVE?

21 MEASUREMENTS

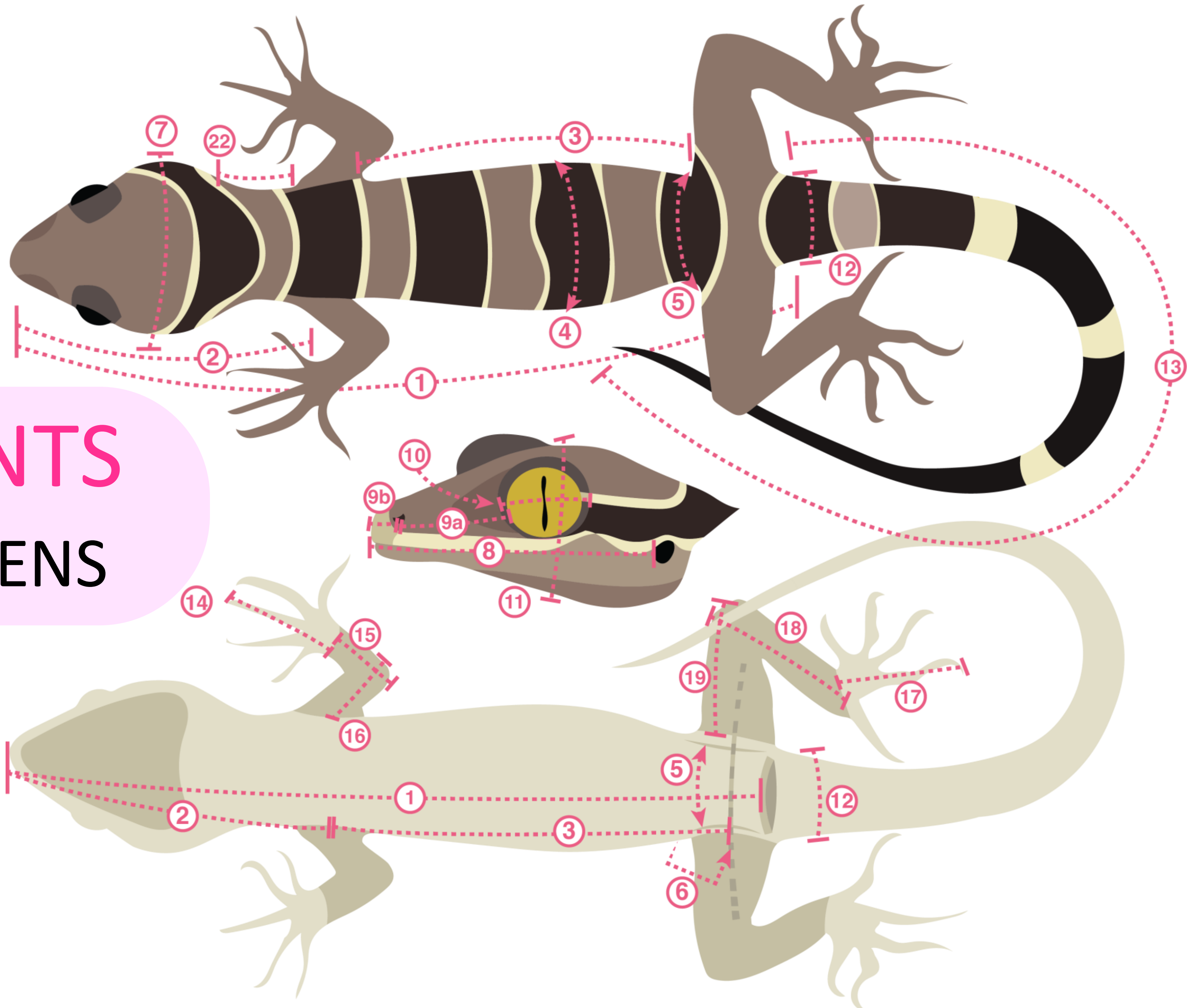
FROM **480+** SPECIMENS

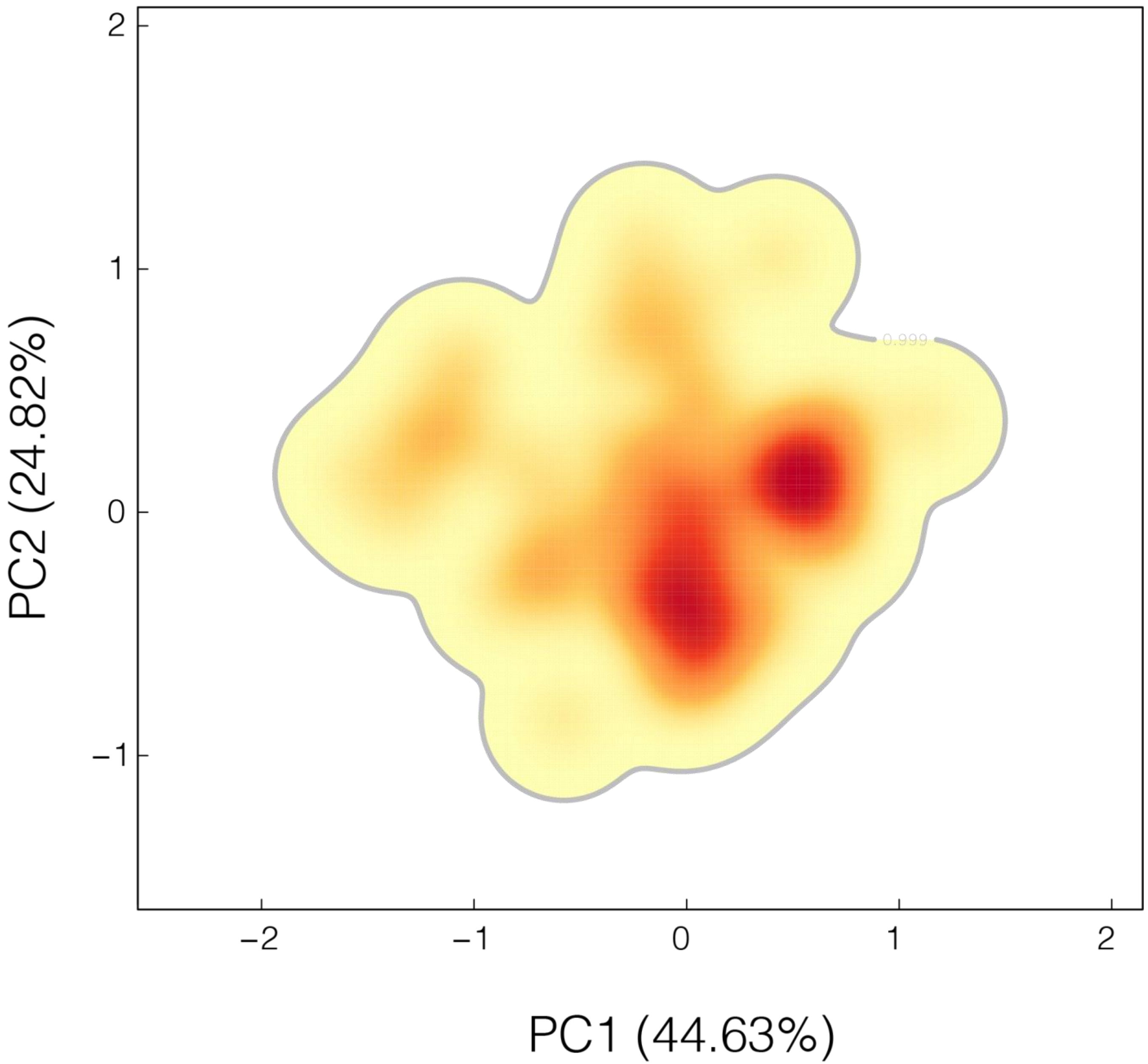


21 MEASUREMENTS

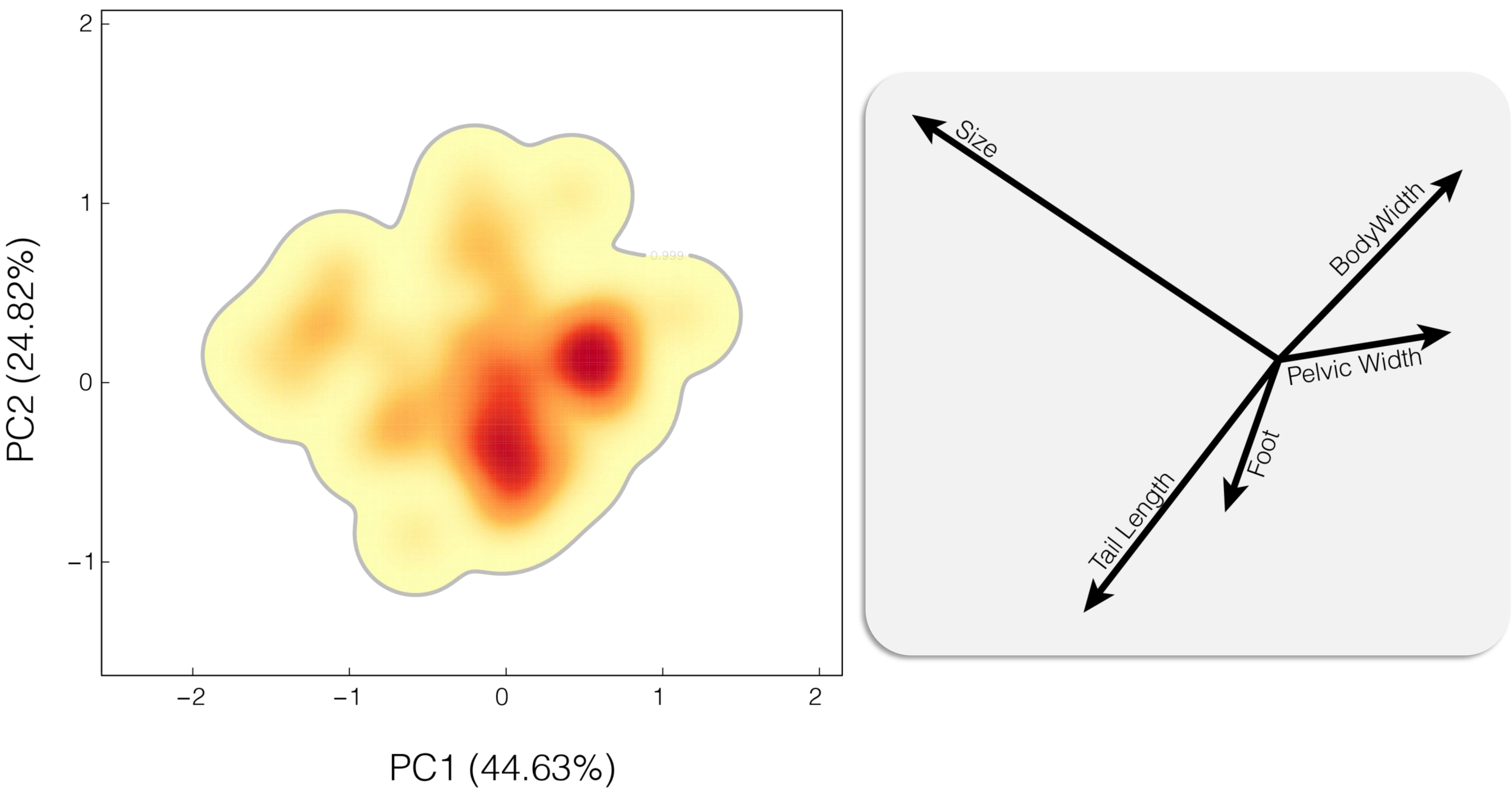
FROM **480+** SPECIMENS

120+ SPECIES

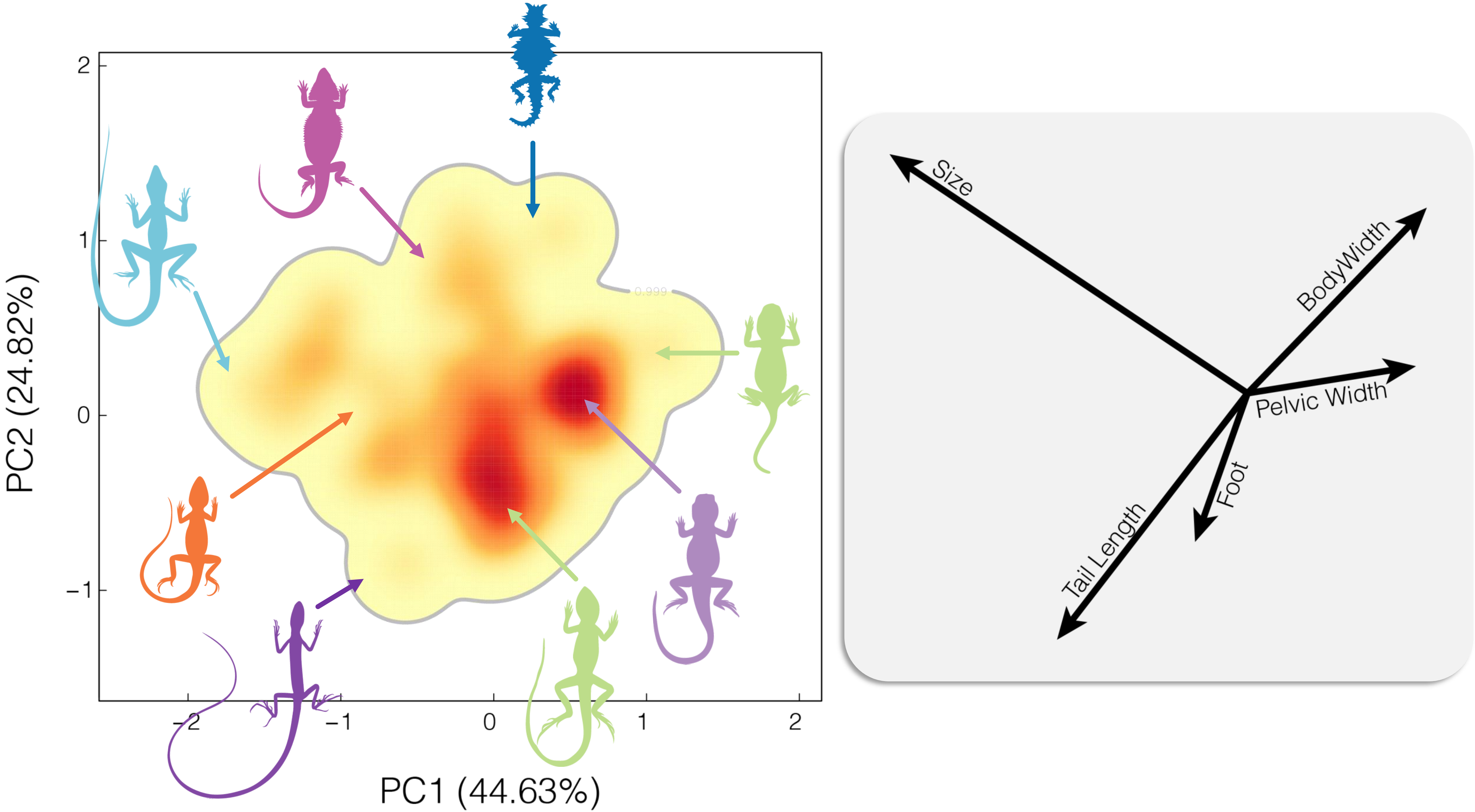




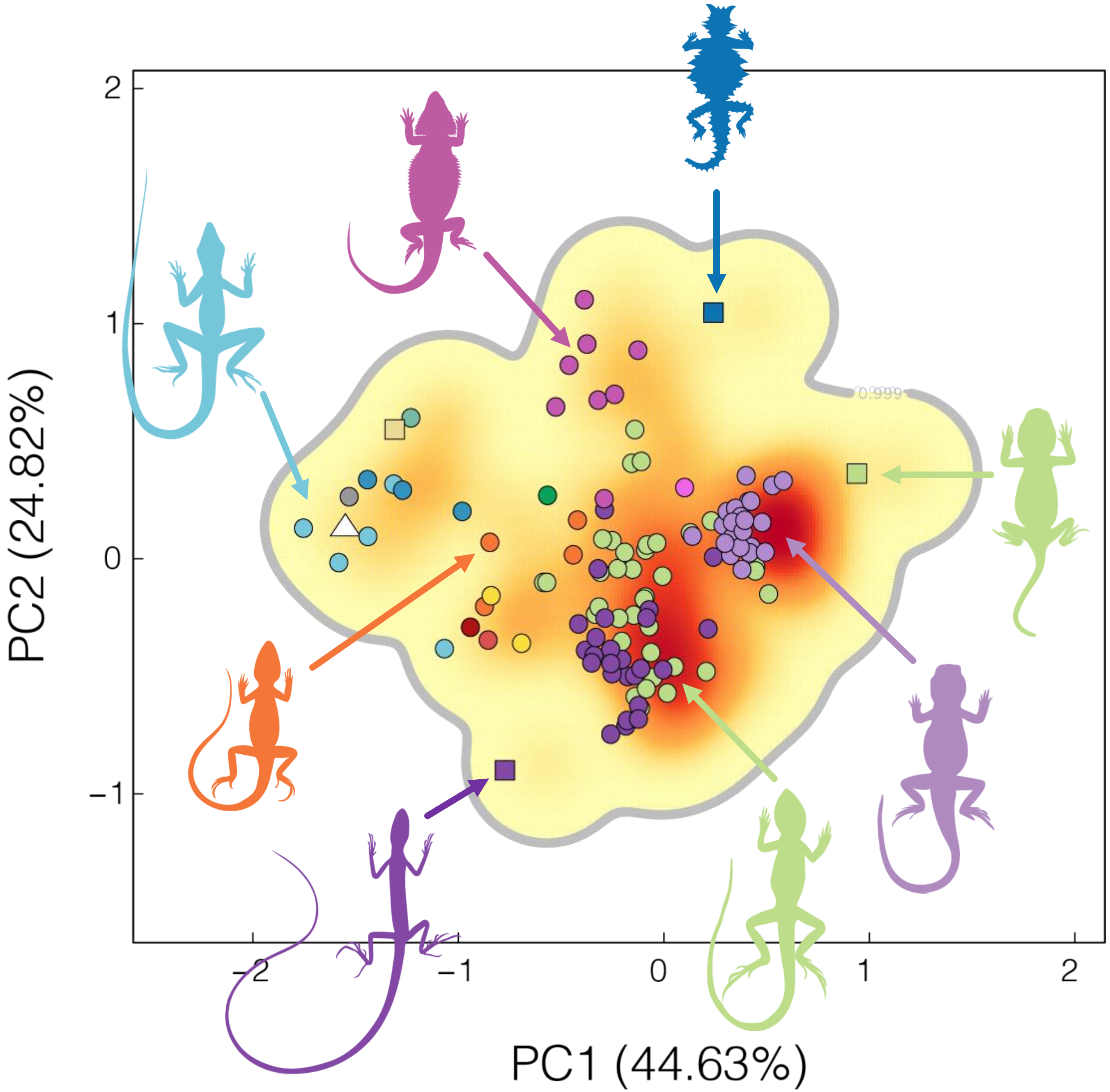
PC2 (24.82%)
PC1 (44.63%)
2
1
0
−1
−2
−1
0
1
2
0.999



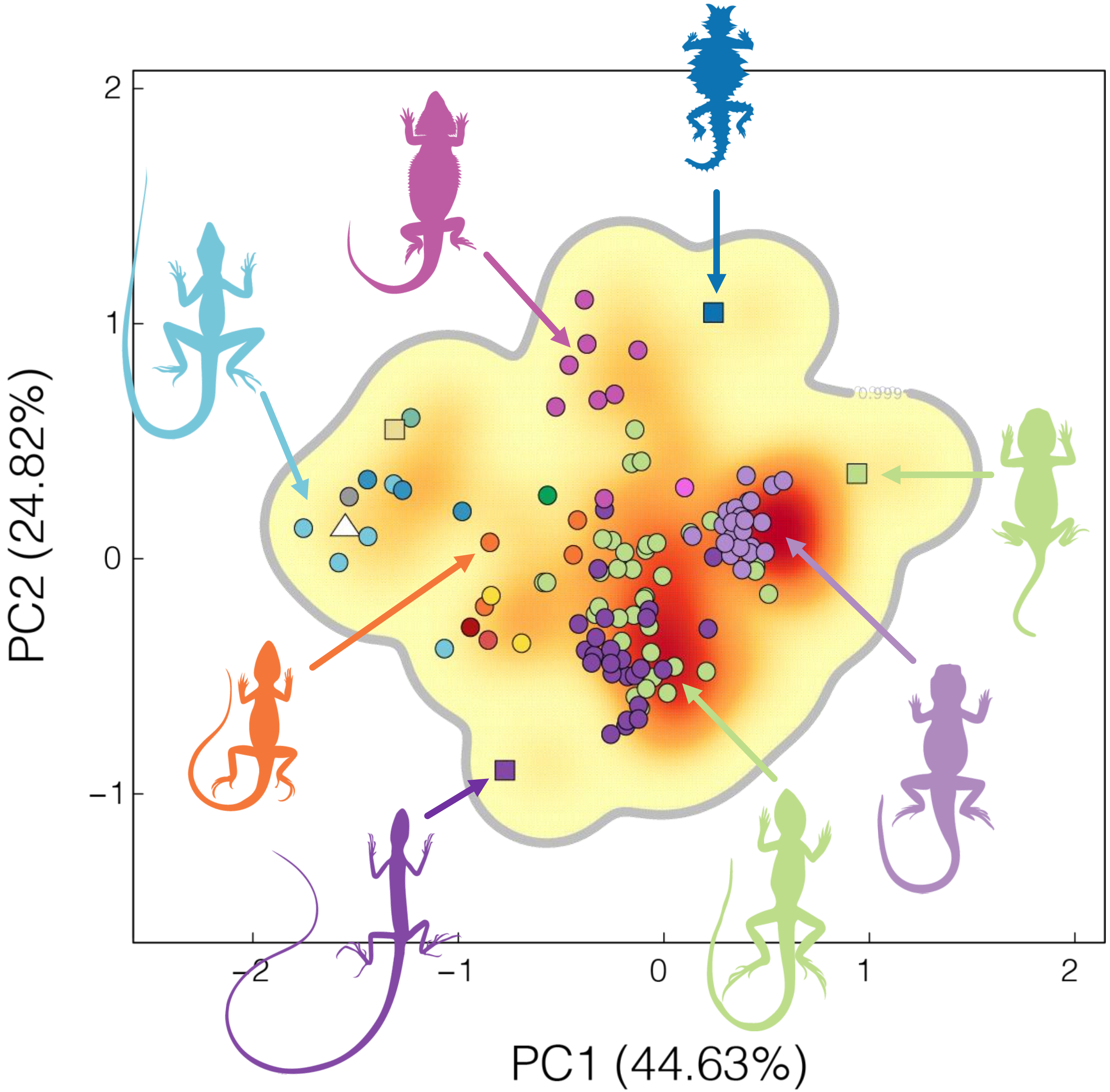
PC2 (24.82%)
PC1 (44.63%)
2
1
0
−1
−2
−1
0
1
2
0.999
Size
BodyWidth
Pelvic Width
Foot
Tail Length



PC2 (24.82%)
PC1 (44.63%)
2
1
0
−1
−2
−1
0
1
2
0.999
Size
BodyWidth
Pelvic Width
Foot
Tail Length

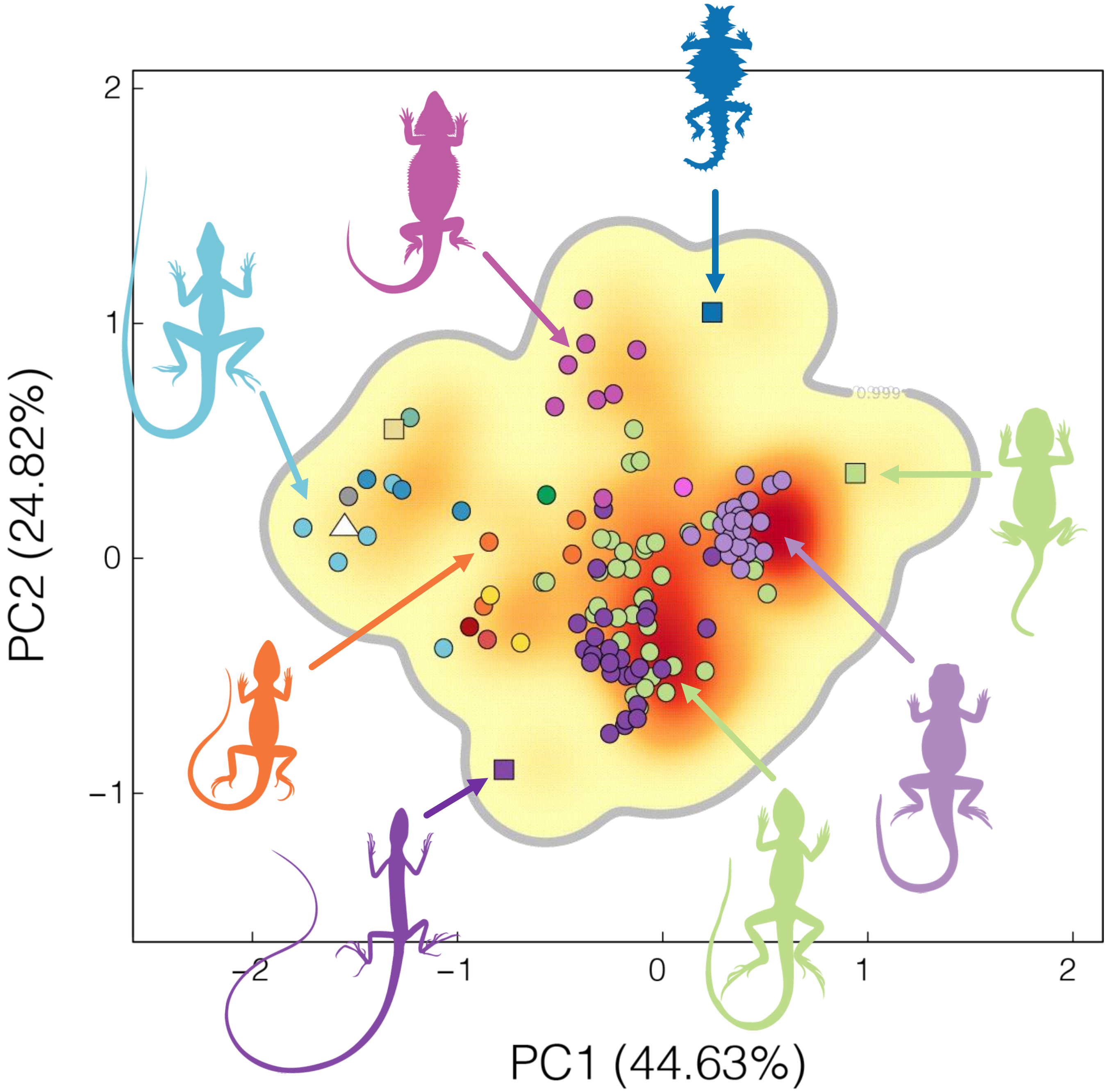


HOW MANY *PEAKS* IN THE LANDSCAPE?



HOW MANY *PEAKS* IN THE LANDSCAPE?

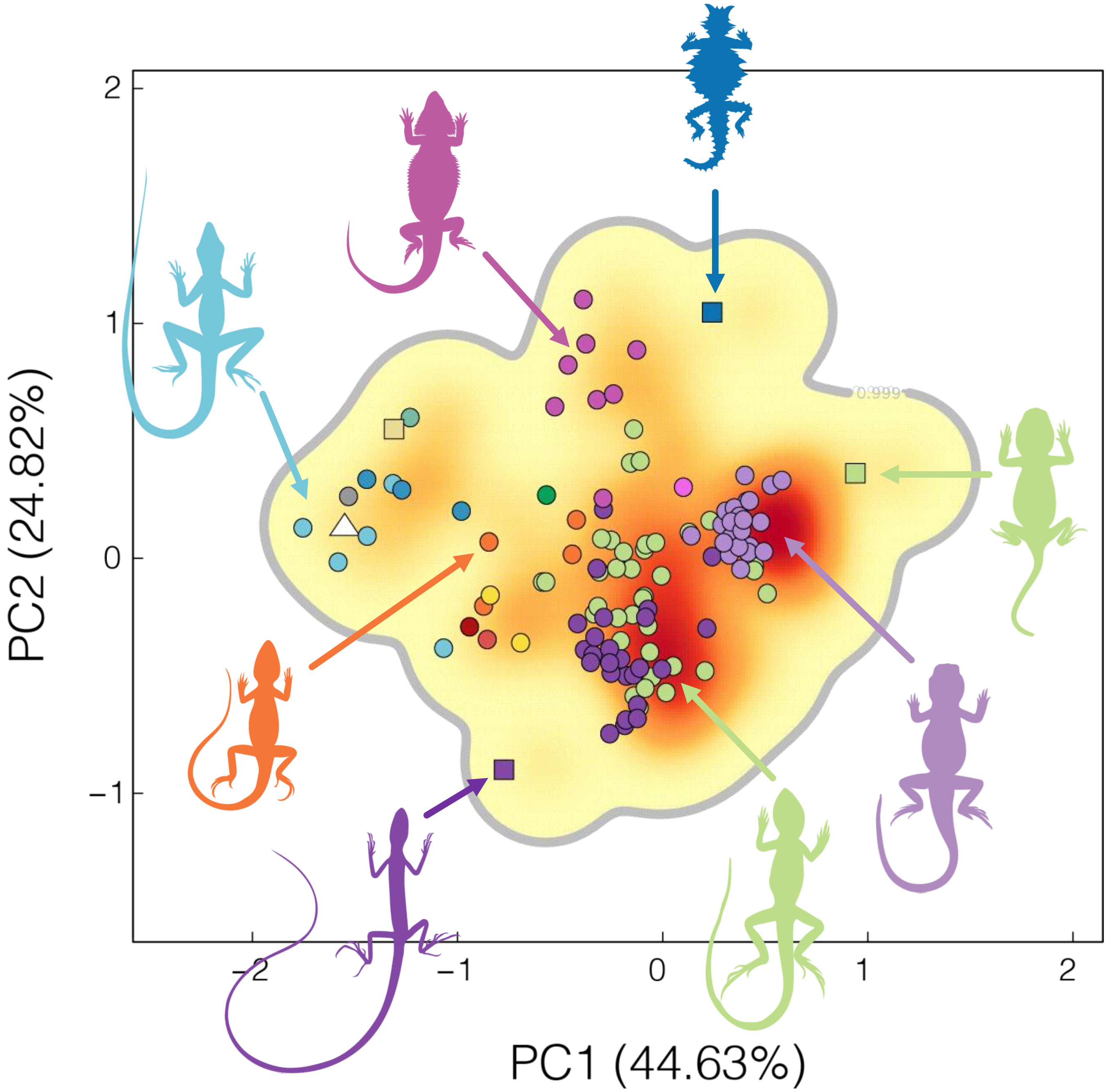
HOW DO *SPECIES* MATCH TO PEAKS?



HOW MANY *PEAKS* IN THE LANDSCAPE?

HOW DO *SPECIES* MATCH TO PEAKS?

DO *ANCESTORS* MATCH TO PEAKS?



HOW MANY *PEAKS* IN THE LANDSCAPE?

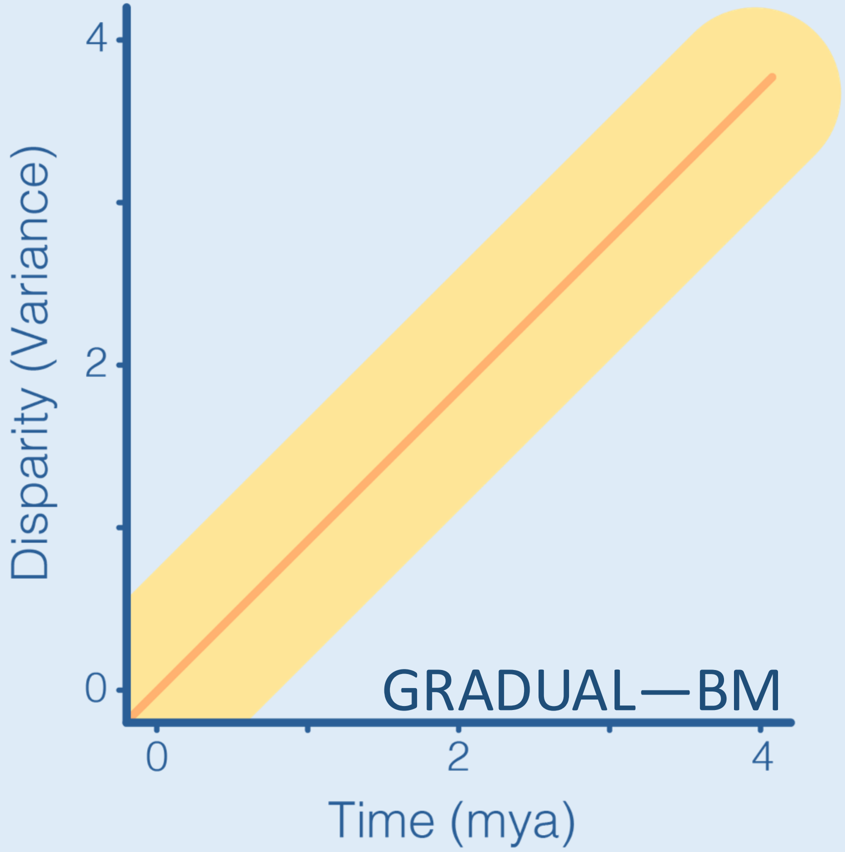
HOW DO *SPECIES* MATCH TO PEAKS?

DO *ANCESTORS* MATCH TO PEAKS?

WHAT'S THE PATH TO *NEW* PEAKS?

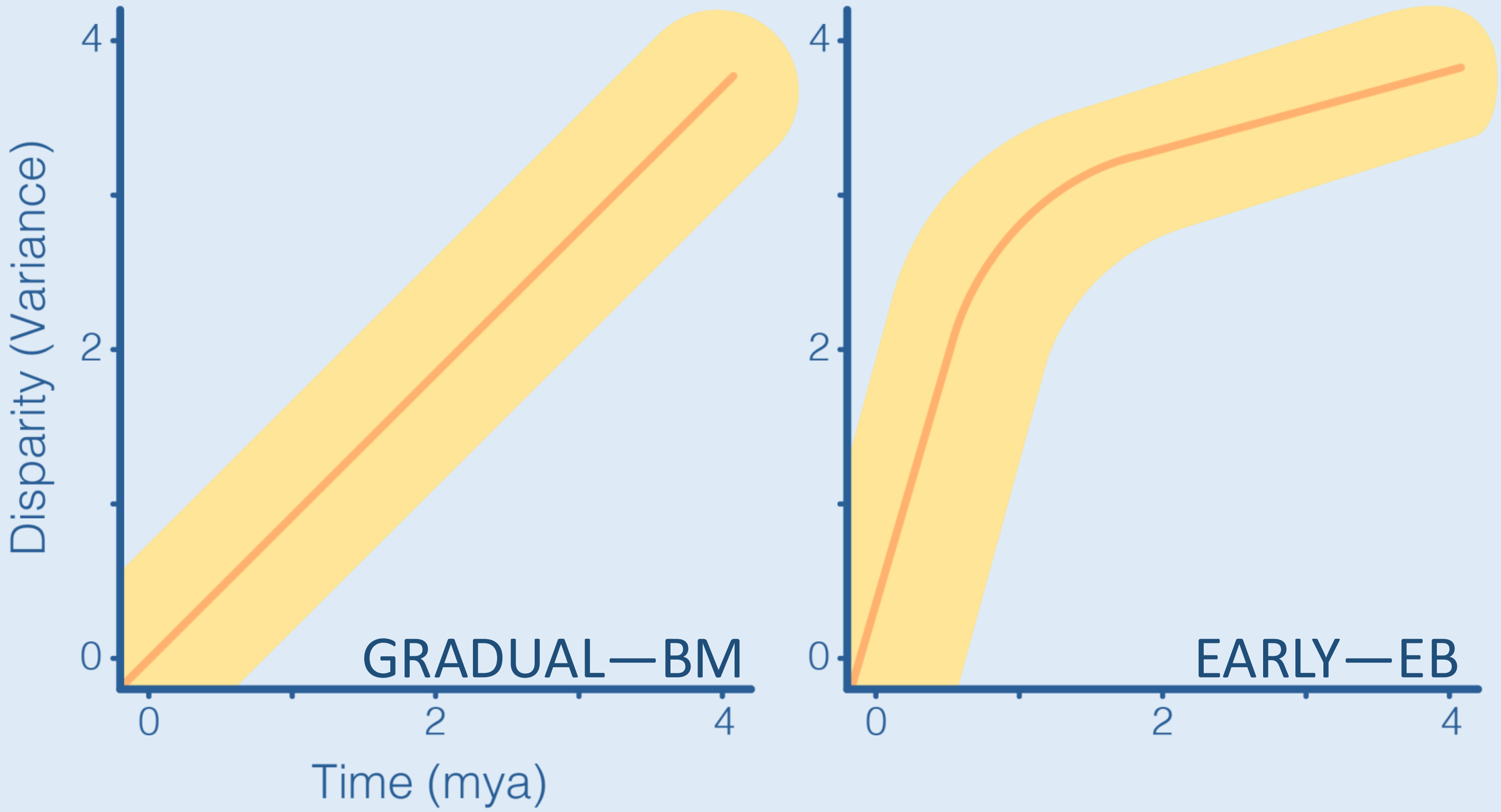
TRAIT MODEL FITTING

SUPPORTS BRANCH-SPECIFIC RATES



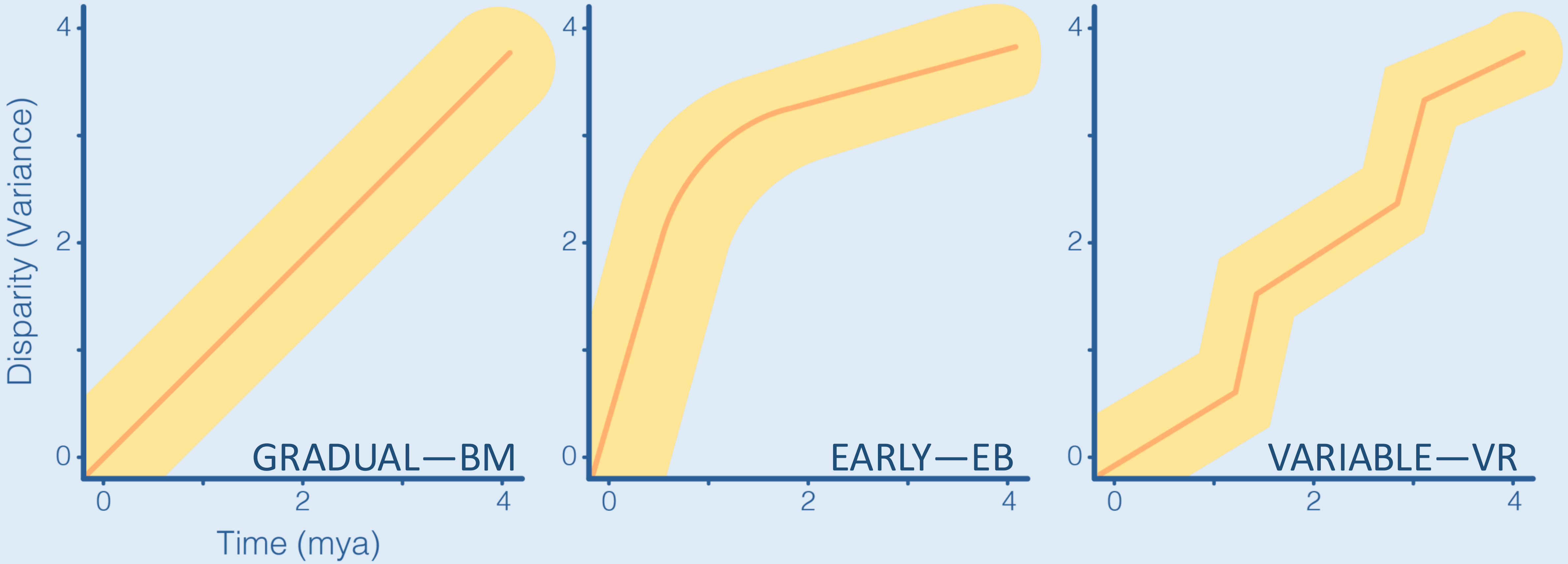
TRAIT MODEL FITTING

SUPPORTS BRANCH-SPECIFIC RATES



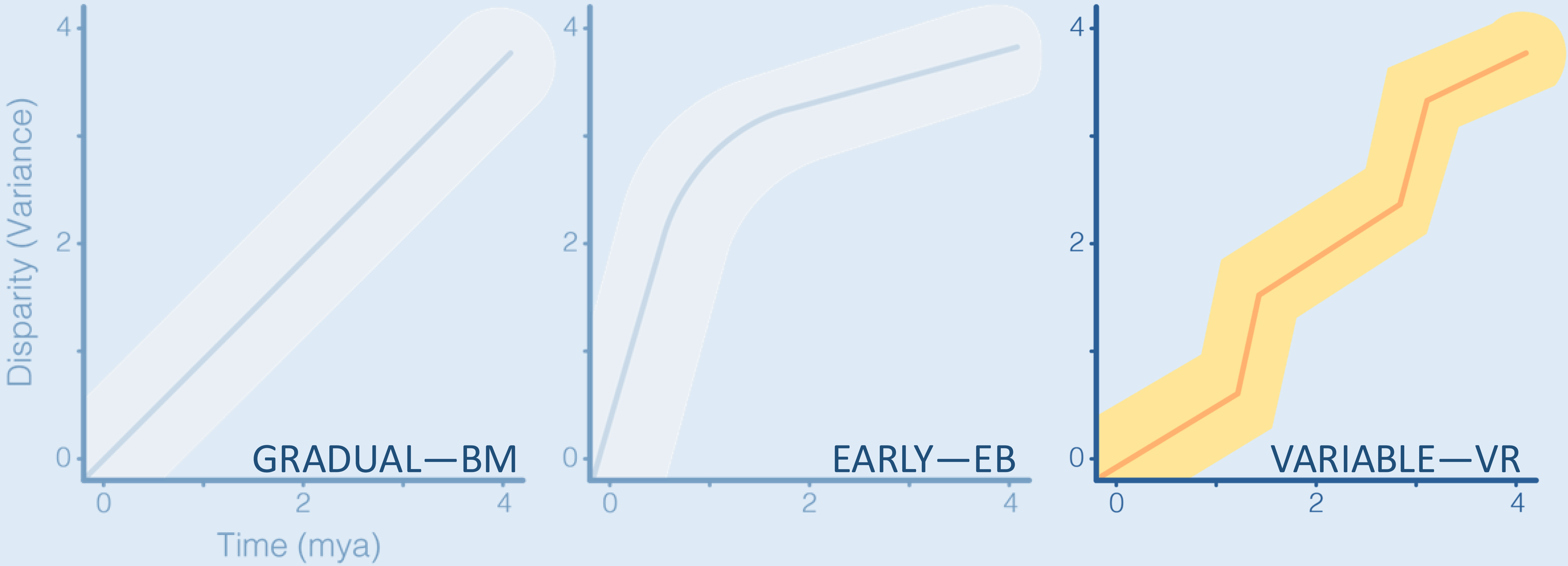
TRAIT MODEL FITTING

SUPPORTS BRANCH-SPECIFIC RATES



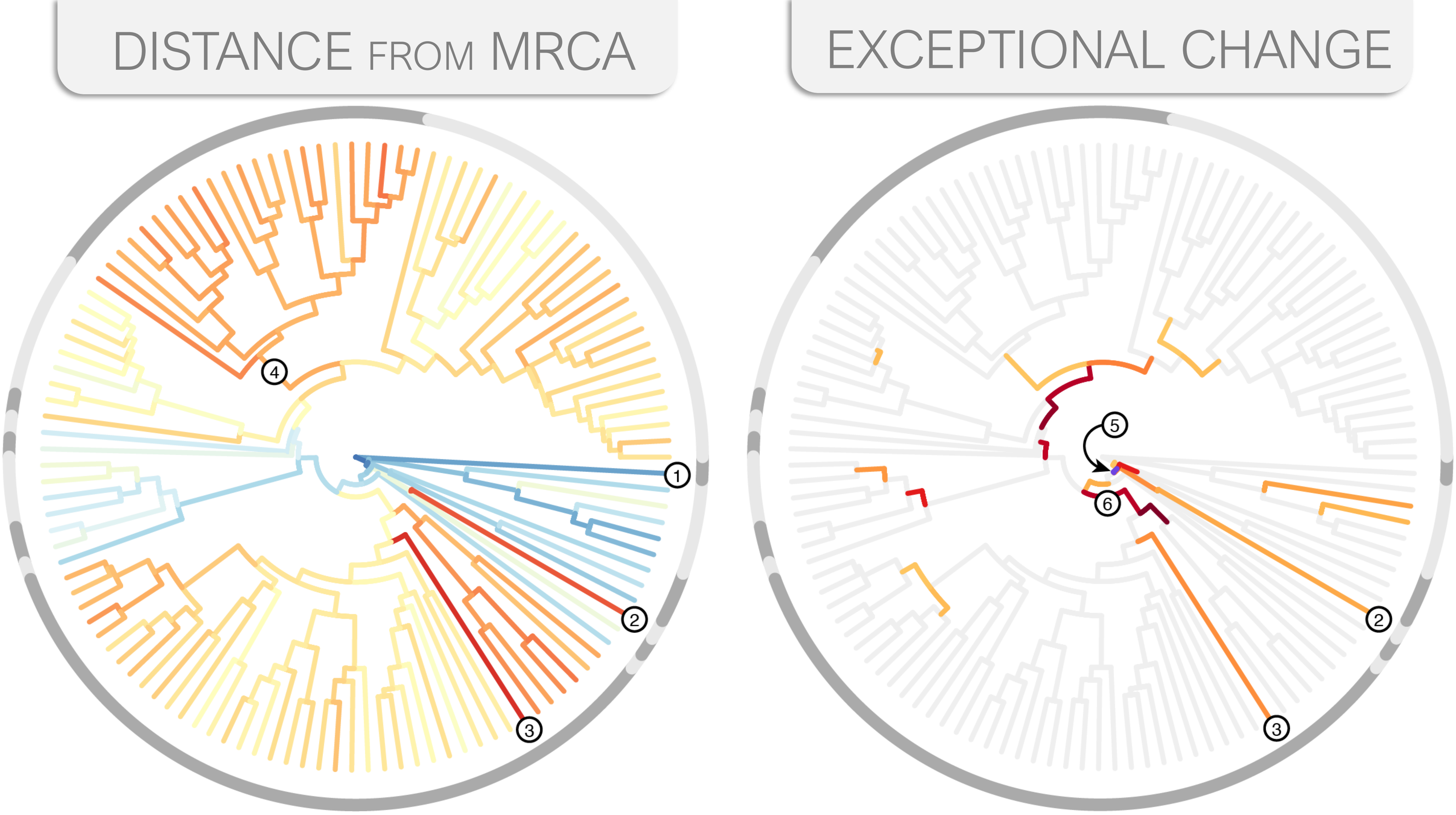
TRAIT MODEL FITTING

SUPPORTS BRANCH-SPECIFIC RATES



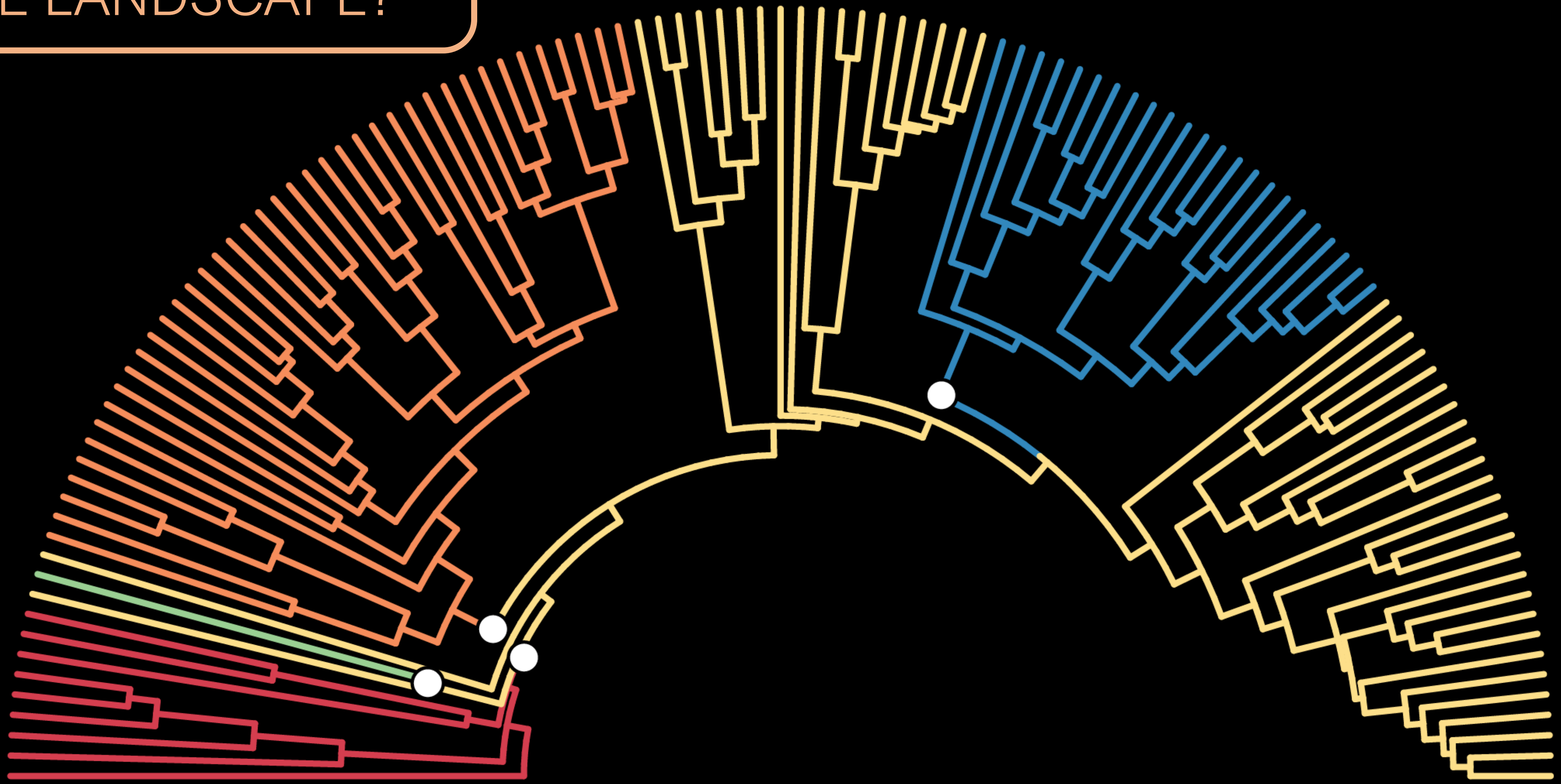
DISTANCE FROM MRCA

1 *Physignathus*　2 *Moloch*　3 *Cryptagama*　4 *Tympanocryptis*　5 MRCA Australia　6 *Ctenophorus*



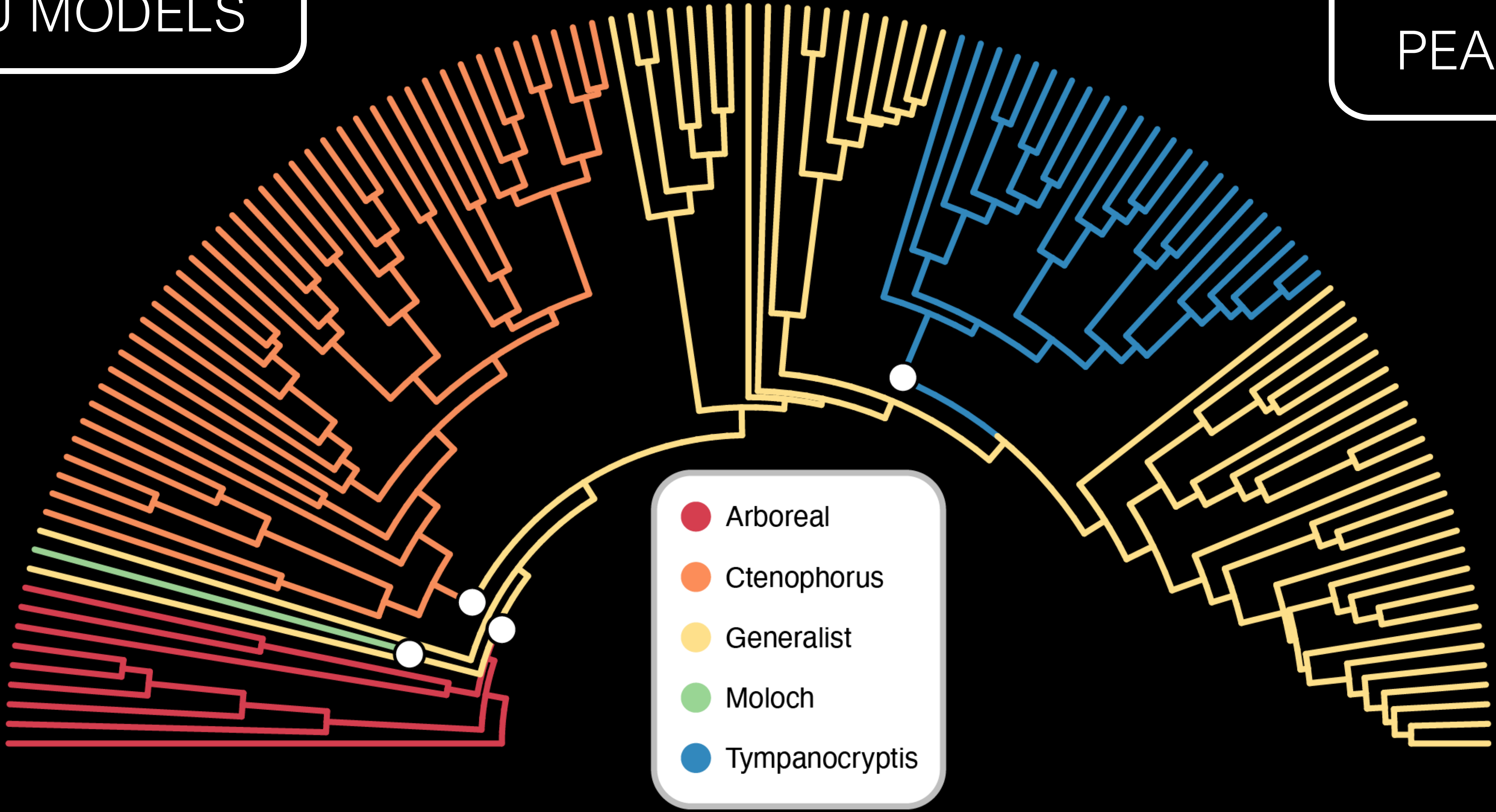
1 *Physignathus* 2 *Moloch* 3 *Cryptagama* 4 *Tympanocryptis* 5 MRCA Australia 6 *Ctenophorus*

HOW MANY *PEAKS* IN THE LANDSCAPE?



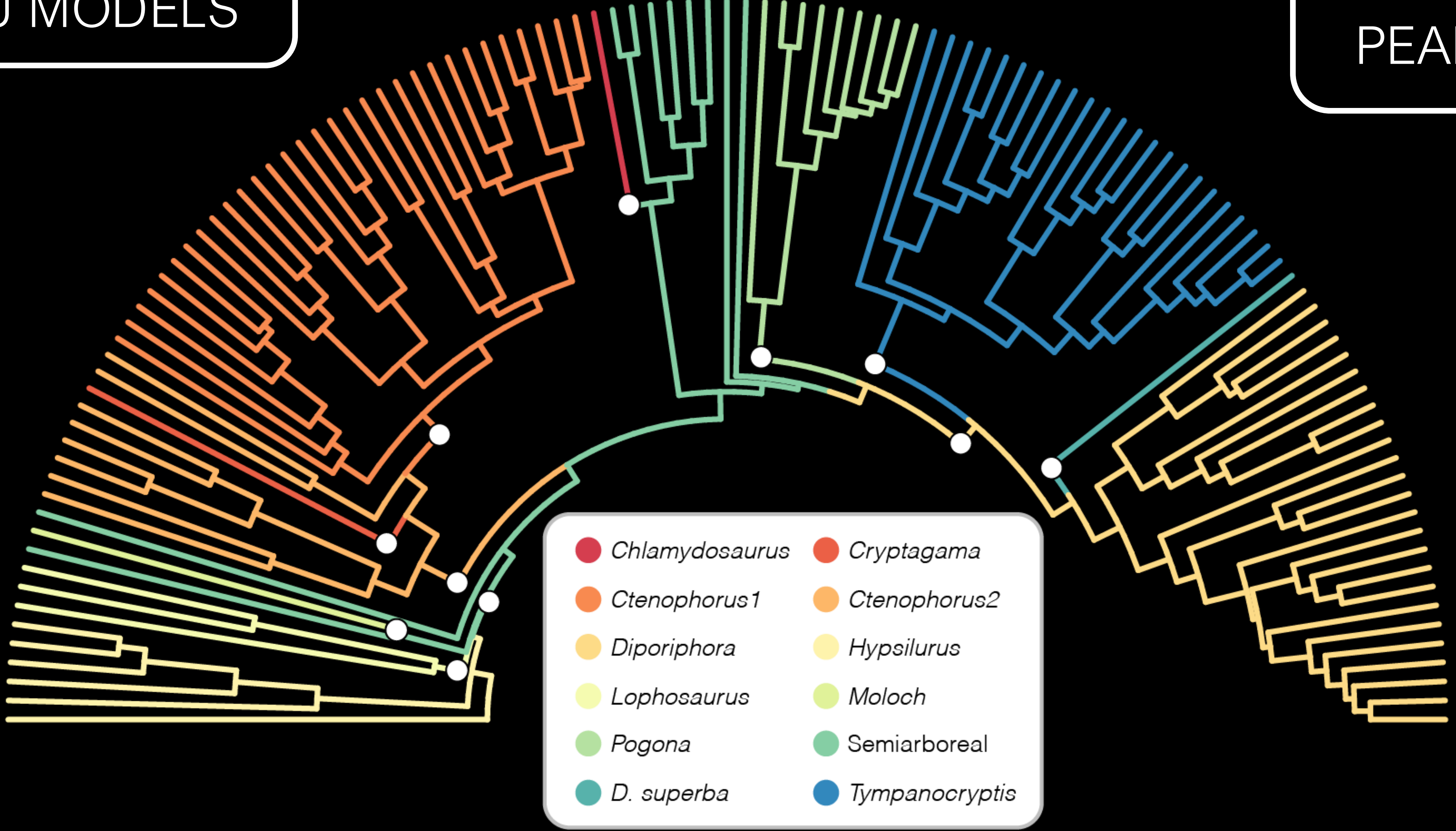
MULTI OPTIMA
OU MODELS

5
PEAKS

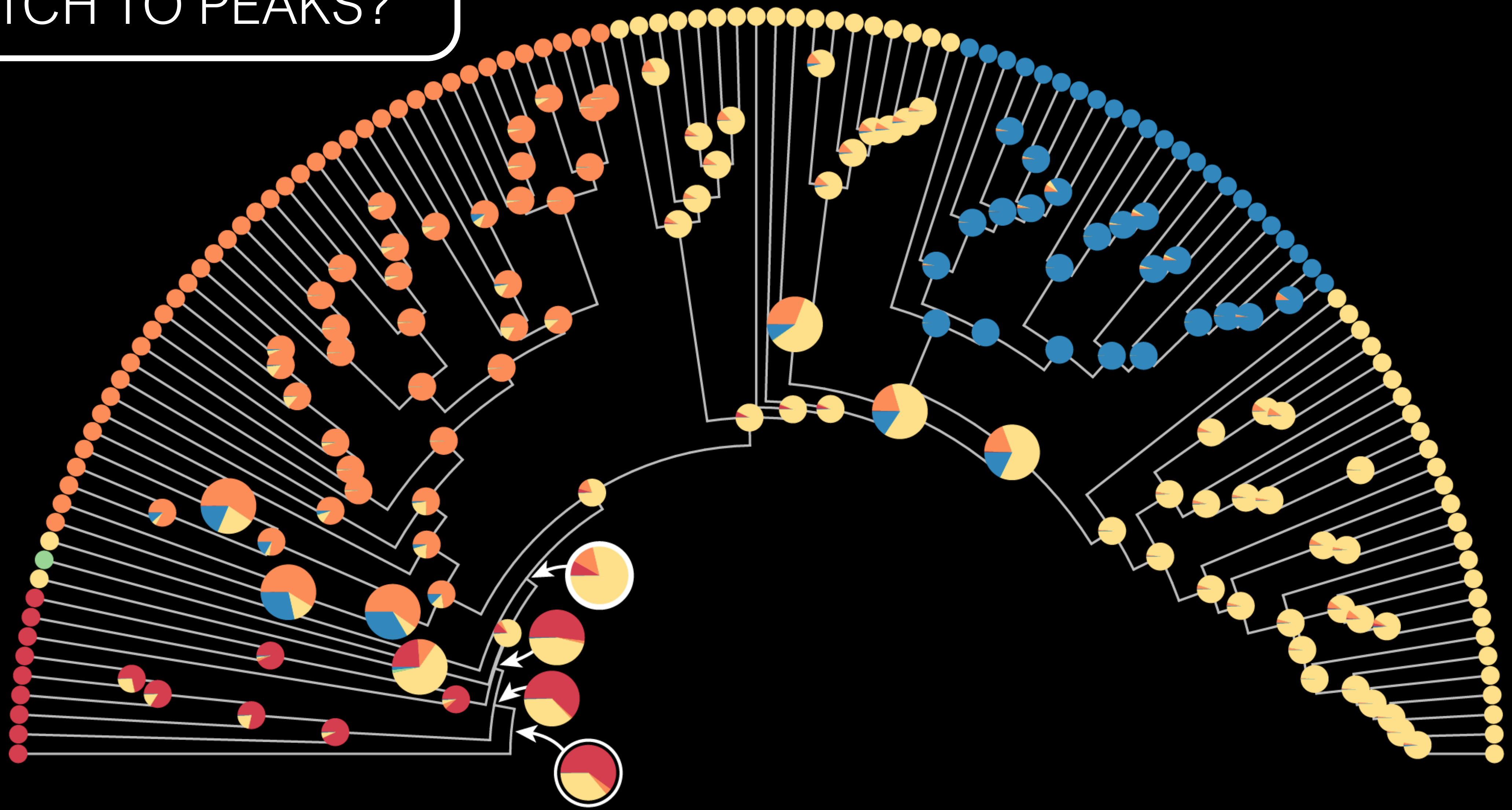


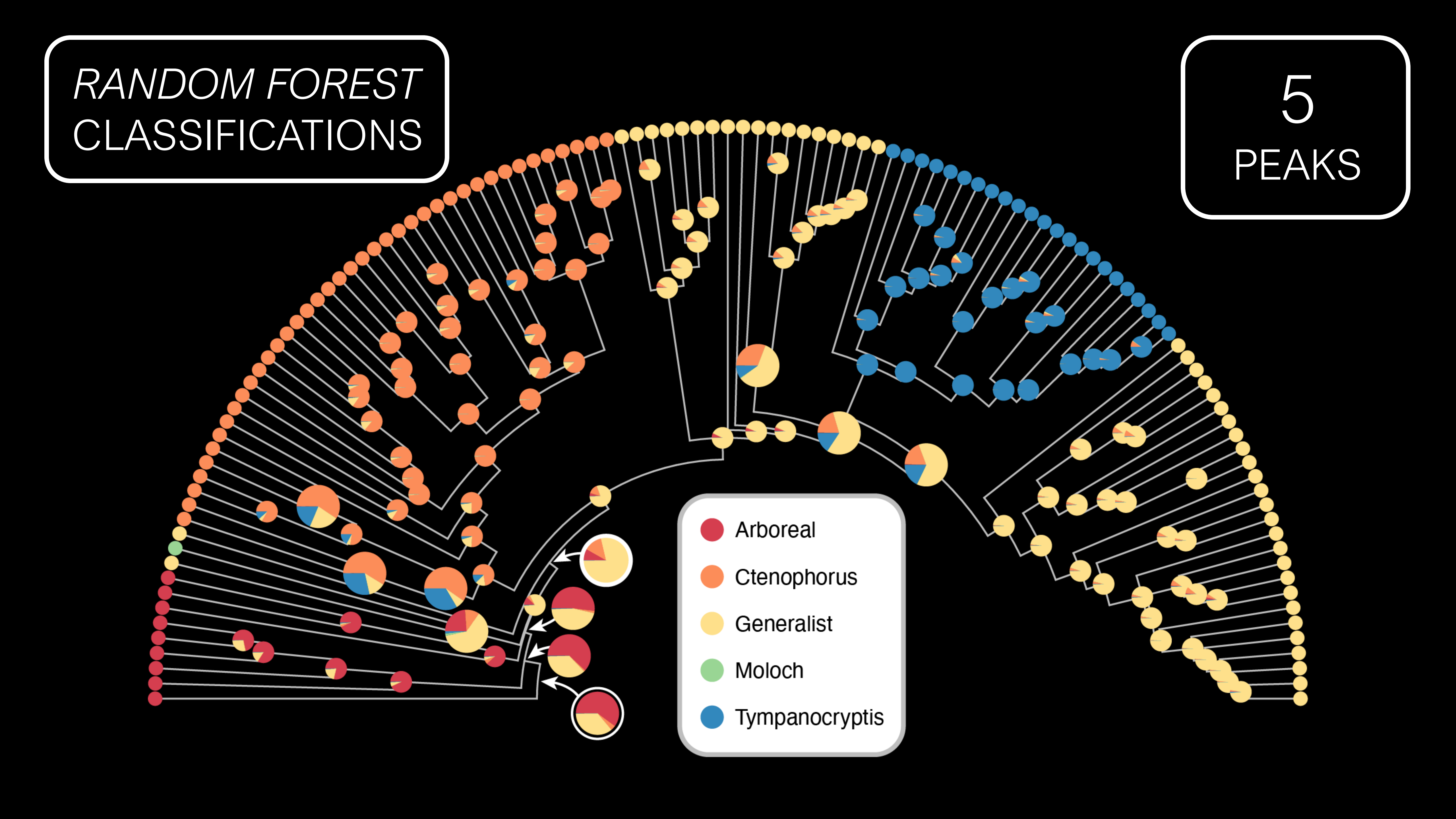
MULTI OPTIMA OU MODELS

12 PEAKS



DO *ANCESTORS* MATCH TO PEAKS?

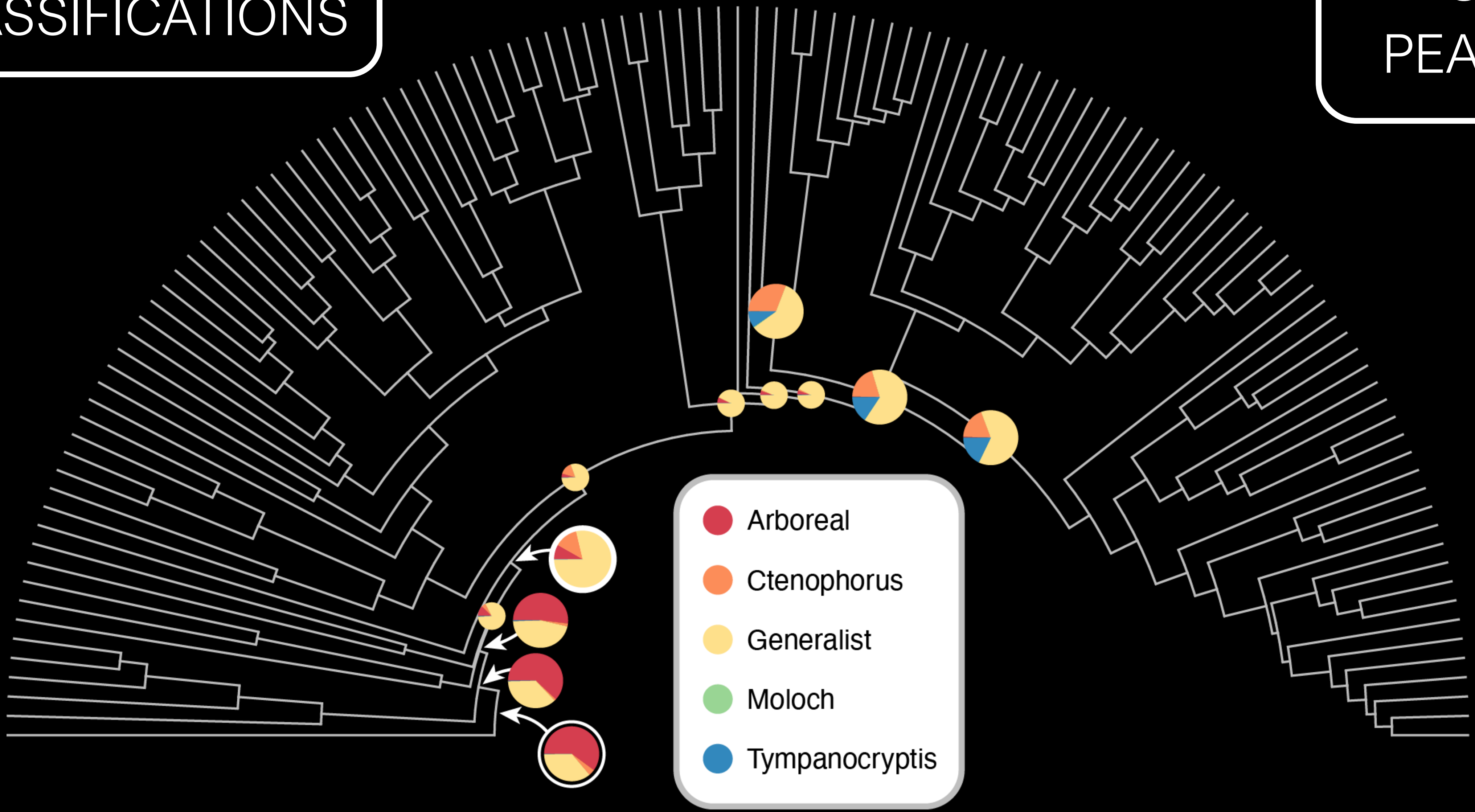




RANDOM FOREST
CLASSIFICATIONS
5
PEAKS
Arboreal
Ctenophorus
Generalist
Moloch
Tympanocryptis

RANDOM FOREST
CLASSIFICATIONS

5
PEAKS



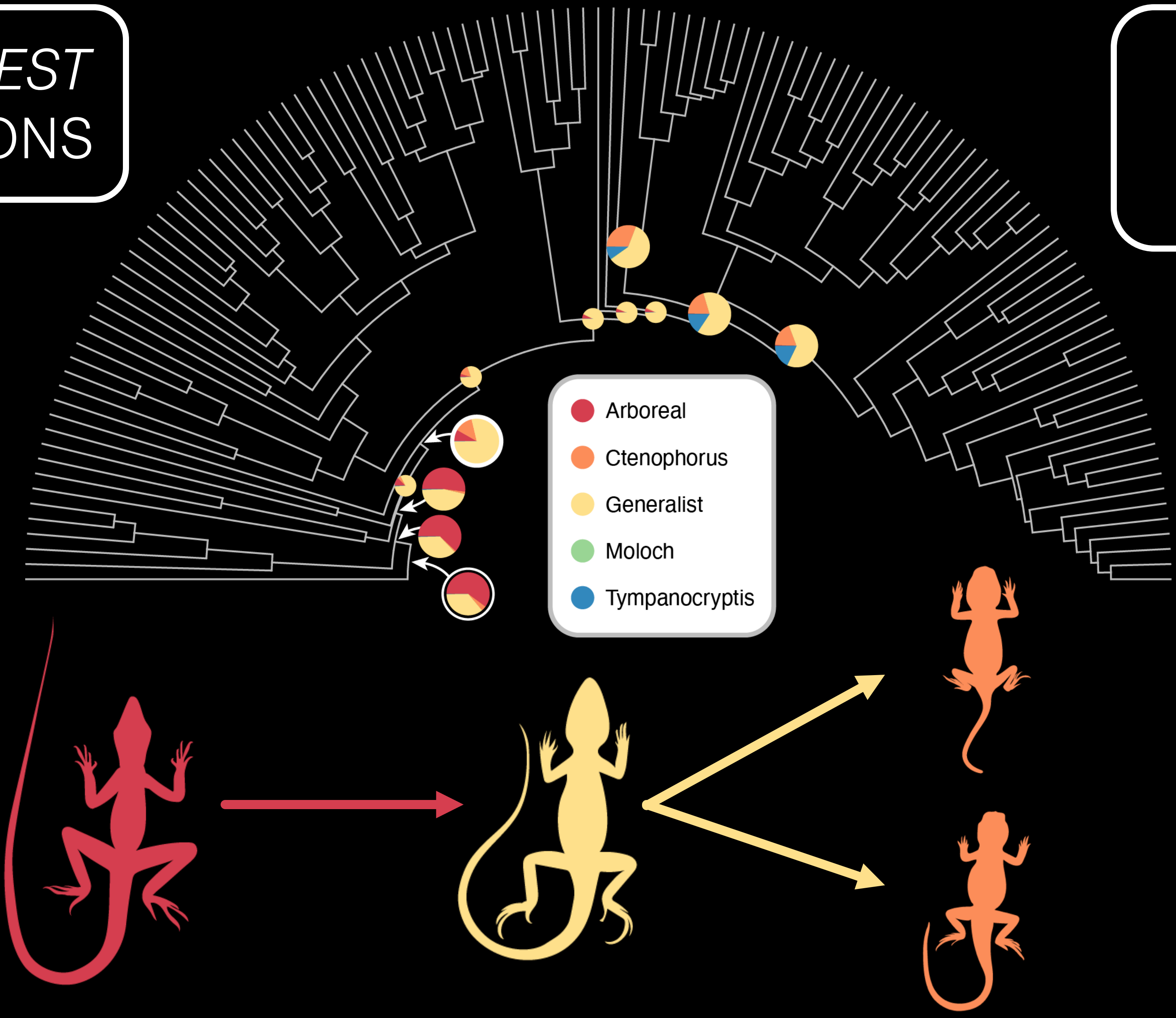
HOW DO WE GET

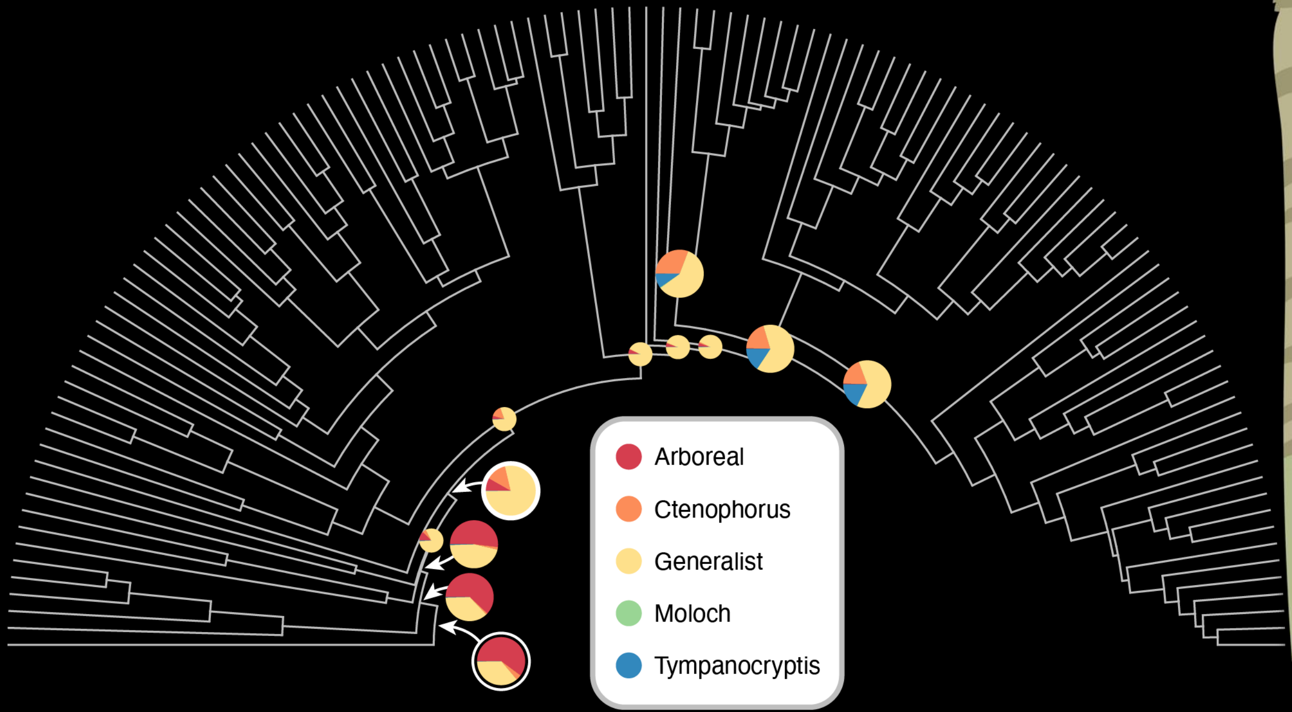
FROM THIS


HOW DO WE GET
FROM THIS → TO THIS
?

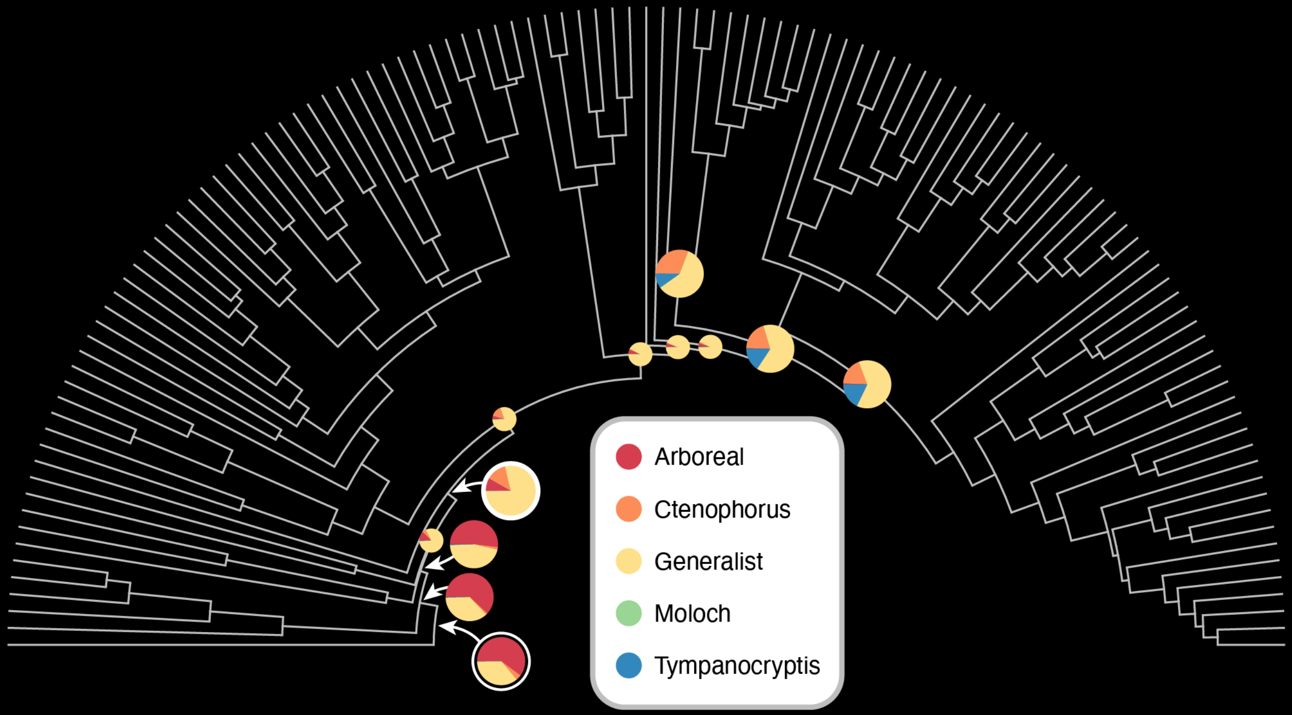
RANDOM FOREST CLASSIFICATIONS

5 PEAKS



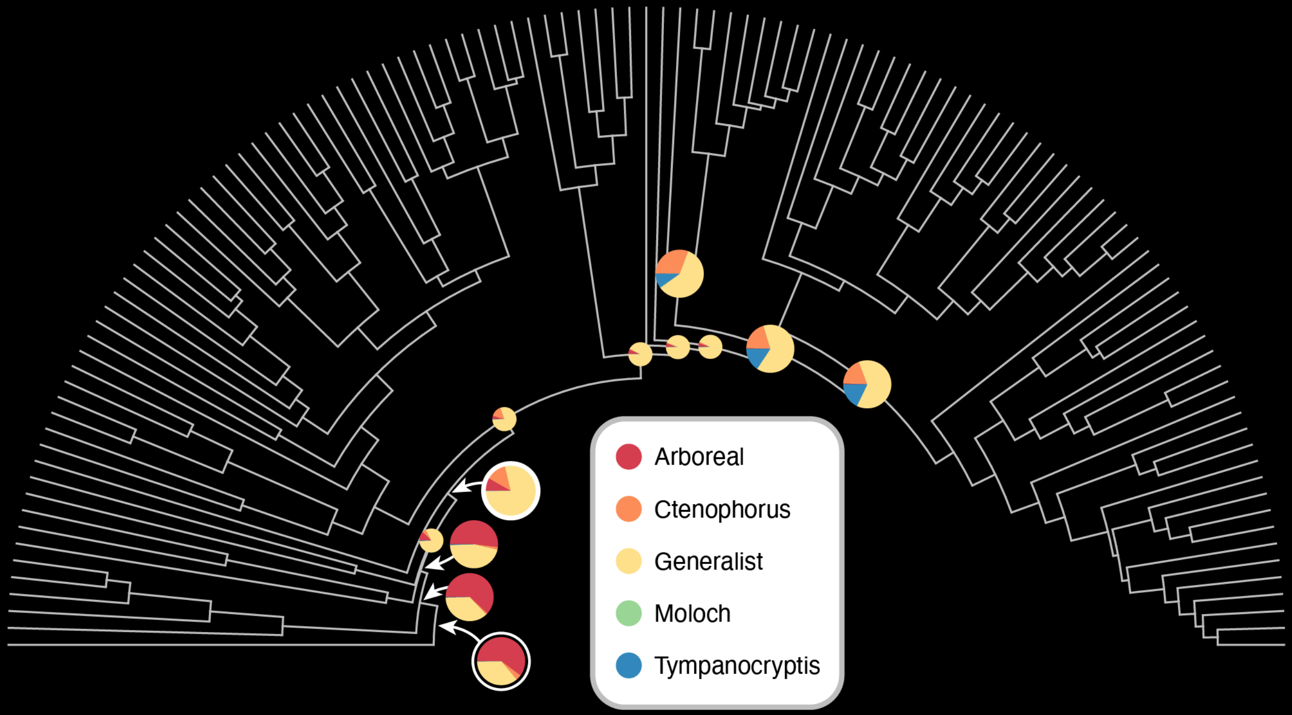


ANCESTRAL DRAGON
WAS *ARBOREAL*



ANCESTRAL DRAGON
WAS *ARBOREAL*

AUSTRALIAN ANCESTOR
WAS *GENERALIST*



ANCESTRAL DRAGON
WAS *ARBOREAL*

AUSTRALIAN ANCESTOR
WAS *GENERALIST*

SPECIALISTS EVOLVE FROM
GENERALIST ANCESTORS

READ MORE HERE

bioRxiv

THE PREPRINT SERVER FOR BIOLOGY

'Generalists link peaks in the shifting adaptive landscape of Australia's dragon lizards'

ANCESTRAL DRAGON WAS *ARBOREAL*

AUSTRALIAN ANCESTOR WAS *GENERALIST*

SPECIALISTS EVOLVE FROM *GENERALIST* ANCESTORS